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$$H_0(x, y) = Gx + F(x^2 + y^2), \quad H_1(\lambda, x, y) = U(x, y) \cos \lambda + V(x, y) \sin \lambda,$$

$$U(x, y) = C(x^2 - y^2) + Dx + E, \quad V(x, y) = 2Cxy + Dy,$$

$$\alpha = -\frac{3(1-\nu)^2}{L_{res}^4} < 0, \quad L_{res} = \left(\frac{(1-\nu)^2}{3} \right)^{1/3}.$$

$$, \quad \varepsilon = \sqrt{\nu} \quad \alpha:$$

$$t \rightarrow \varepsilon t, \quad \Lambda \rightarrow (\varepsilon e_J^2)^{-1} \Lambda, \quad x \rightarrow e_J^{-1} x, \quad y \rightarrow e_J^{-1} y, \quad \alpha \rightarrow e_J^2 \alpha$$

()

$$H_w = \alpha \frac{\Lambda^2}{2} + H_0(x, y) + H_1(\lambda, x, y), \quad (2)$$

ε

$$\Omega_w = d\Lambda \wedge d\lambda + \varepsilon^{-1} dx \wedge dy. \quad (3)$$

$$\begin{aligned} & ; \quad (!!![6]) \quad . \quad 2\pi - H_w \quad \lambda, \quad S^1 \times \mathbb{R}^3. \\ & , \quad [6] \quad , \quad H_w \quad e_J. \quad , \quad (e_J \rightarrow 0+) \quad , \quad . \\ & , \quad . \quad , \quad \alpha, B, C, D, E, F \quad , \end{aligned}$$

1. , C, D, E, F, G :

$$D^2 - 4CE > 0, \quad (4a)$$

$$C > |F|, \quad (4b)$$

$$(DF - GC)^2 < (D^2 - 4CE)(|F| - C)^2. \quad (4c)$$

.

3. -

$$H_w \text{ (2)} \quad \Omega_w \text{ (3)}$$

$$\begin{cases} \dot{\Lambda} = U \sin \lambda - V \cos \lambda, & \dot{\lambda} = \alpha \Lambda, \\ \dot{x} = -\varepsilon(2Fy + \frac{\partial U}{\partial y} \cos \lambda + \frac{\partial V}{\partial y} \sin \lambda), & \dot{y} = \varepsilon(2Fx + G + \frac{\partial U}{\partial x} \cos \lambda + \frac{\partial V}{\partial x} \sin \lambda). \end{cases} \quad (5)$$

$$\begin{aligned} & - \quad (\Lambda, \lambda) \quad (x, y). \\ & . \quad . \quad . \quad . \quad !!![8], !!![7], \quad . \quad , \quad . \quad , \quad : \end{aligned}$$

$$\dot{\xi} = \mathcal{F}(\xi, \eta), \quad \dot{\eta} = \varepsilon \mathcal{G}(\xi, \eta), \quad (6)$$

$$\xi \in \mathbb{R}^n - , \quad \eta \in \mathbb{R}^m - , \quad \varepsilon - . \\ \varepsilon > 0 \quad \tau = \varepsilon t. \quad , \quad \tau,$$

$$\varepsilon \xi' = \mathcal{F}(\xi, \eta), \quad \eta' = \mathcal{G}(\xi, \eta). \quad (7)$$

$$\begin{aligned} & (6) \text{ (7)} \quad \varepsilon = 0, \quad . \quad , \quad (6) \text{ (7)} \quad , \quad . \quad , \quad (6) \quad \varepsilon = 0 \quad , \quad , \quad \eta. \quad , \quad \varepsilon = 0 \text{ (7)} \quad \eta' = G(\xi(\eta), \eta), \\ & \xi(\eta) \quad \mathcal{F}(\xi, \eta) = 0. \quad (\xi, \eta), \quad , \quad . \\ & , \quad (6) \text{ (7)}, \quad . \quad , \quad . \end{aligned}$$

3.1.

(5) ε , " " :

$$\begin{cases} \dot{\Lambda} = U(x, y) \sin \lambda - V(x, y) \cos \lambda, & \dot{\lambda} = \alpha \Lambda, \\ \dot{x} = 0, & \dot{y} = 0. \end{cases} \quad (8)$$

$(x, y) \in U, V \neq 0$.
 $(x, y) \in U(x, y) \neq 0, V(x, y) \neq 0$, (8) .
 $\lambda_{\pm} : \Pi_0 \rightarrow S^1, \beta : \mathbb{R}^2 \rightarrow S^1$:

$$\Pi_0 \equiv \mathbb{R}^2 \setminus \{(x, y) \mid U(x, y) = 0, V(x, y) = 0\}$$

$$\lambda_+(x, y) \equiv \begin{cases} \arctan \frac{V}{U}, & U > 0, V > 0, \\ \arctan \frac{V}{U} + \pi, & U < 0, \\ \arctan \frac{V}{U} + 2\pi, & U > 0, V < 0, \end{cases} \quad \lambda_-(x, y) \equiv \begin{cases} \arctan \frac{V}{U} + \pi, & U > 0, V > 0, \\ \arctan \frac{V}{U} + 2\pi, & U < 0, \\ \arctan \frac{V}{U} + 3\pi, & U > 0, V < 0. \end{cases}$$

$\lambda = t - \Lambda$:

$$\ddot{\lambda} - \alpha U \sin \lambda + \alpha V \cos \lambda = 0,$$

($\alpha < 0$) :

$$\ddot{\lambda} - \alpha \beta \sin(\lambda - \lambda_+(x, y)) = 0, \quad \beta(x, y) \equiv \sqrt{U^2 + V^2} > 0.$$

$$U = V = 0, \quad U(x, y) = 0, \quad V(x, y) = 0 : \quad (4b)$$

$$(x_b^{\pm}, y_b^{\pm}) \equiv \left(\frac{-D \pm \sqrt{D^2 - 4EC}}{2C}, 0 \right)$$

$$\Pi_0 = \mathbb{R}^2 \setminus \{(x_b^{\pm}, y_b^{\pm})\} .$$

$$(\Lambda, \lambda) = (\lambda_+(x, y), 0) \quad (\Lambda, \lambda) = (\lambda_-(x, y), 0),$$

$$\lambda_{sep}(t, t_0) = \lambda_+(x, y) + 2 \arctan \sinh(\sqrt{|\alpha|\beta}(t - t_0)), \quad \Lambda(t, t_0) = \frac{\pm 2\beta/|\alpha|}{\cosh(\sqrt{\beta}(t - t_0))}, \quad (9)$$

$$(x, y) = (x_b^{\pm}, y_b^{\pm}) :$$

$$\Lambda = \text{const}, \quad \lambda = \alpha \Lambda t + \text{const}.$$

3.2.

, " " :

$$M_0 = \{(x, y, \lambda, \Lambda) \in S^1 \times \mathbb{R}^3 \mid \Lambda = 0, \quad u(x, y) \sin \lambda = v(x, y) \cos \lambda\}. \quad (10)$$

$$(x, y) \in \Pi_0 \quad (10) \quad \lambda = \lambda_{\pm}(x, y). \quad (x, y) = (x_b^{\pm}, y_b^{\pm}) \quad (10) \quad \lambda \in S^1.$$

$$M_{0,\pm}, \quad M_{0,b}^{\pm}:$$

$$M_0 = M_{0,+} \cup M_{0,-} \cup M_{0,b}^+ \cup M_{0,b}^-,$$

$$M_{0,\pm} = \{\lambda = \lambda_{\pm}(x, y), \Lambda = 0, (x, y) \in \mathbb{R}^2 \setminus \{U = V = 0\}\},$$

$$M_{0,b}^{\pm} = \{y = 0, x = x_b^{\pm}, \Lambda = 0, \lambda \in S^1\}.$$

$$M_{0,\pm} \subset \Pi_0, \quad M_{0,b}^{\pm} \subset S^1.$$

$$M_{0,+} \cap M_{0,-} = \emptyset, \quad M_{0,+} \cap M_{0,b}^{\pm} = \emptyset, \quad M_{0,-} \cap M_{0,b}^{\pm} = \emptyset.$$

3.3.

$\tau = \varepsilon t$ (5), (τ):

$$\begin{cases} \varepsilon \Lambda' = U \sin \lambda - V \cos \lambda, & \varepsilon \lambda' = \alpha \Lambda, \\ x' = -(2Fy + \frac{\partial U}{\partial y} \cos \lambda + \frac{\partial V}{\partial y} \sin \lambda), & y' = 2Fx + G + \frac{\partial U}{\partial x} \cos \lambda + \frac{\partial V}{\partial x} \sin \lambda. \end{cases} \quad (11)$$

ε , " " :

$$\begin{cases} 0 = U \sin \lambda - V \cos \lambda, & 0 = \alpha \Lambda, \\ x' = -(2Fy + \frac{\partial U}{\partial y} \cos \lambda + \frac{\partial V}{\partial y} \sin \lambda), & y' = 2Fx + G + \frac{\partial U}{\partial x} \cos \lambda + \frac{\partial V}{\partial x} \sin \lambda. \end{cases} \quad (12)$$

(12), :

$$x' = -\left(2Fy \pm \frac{\partial \beta}{\partial y}\right), \quad y' = 2Fx + G \pm \frac{\partial \beta}{\partial x}.$$

() $M_{0,+}$, $M_{0,-}$.
() :

$$H_S^\pm = F(x^2 + y^2) + Gx \pm \beta(x, y),$$

,

$$(\hat{x}_0^\pm, 0), \quad \hat{x}_0^\pm = \frac{\pm D - G}{2(F \mp C)}.$$

, , $+\xi_0^\pm, -\xi_0^\pm$ (M_0),

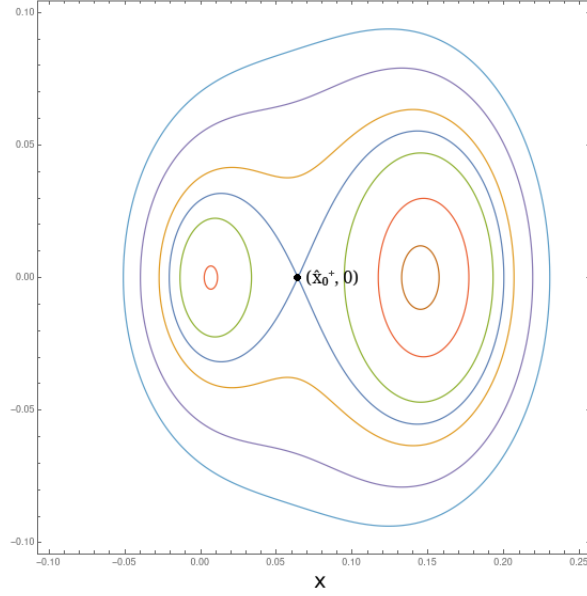
$$\xi_0^\pm = \sqrt{4(C^2 - F^2) + 2\frac{(2C\hat{x}_0^\pm + D)^2}{U(\hat{x}_0^\pm, 0)}(\pm F - C)} \in \mathbb{R}, \quad (13)$$

, (4a) (4c) 1, .

, , () , , $\tau \rightarrow \pm\infty$. , :

$$H_S(x, y) = H_S(x_0^\pm, 0),$$

$$H_S(x_0^\pm, 0) = F((x_0^\pm)^2) + Gx_0^\pm \pm \beta(x_0^\pm, 0).$$



$$. \quad 1. \quad H_S \quad M_{0,+} \cdot \quad M_{0,-}$$

:

$$A = 2U(\hat{x}_0^\pm)(\pm F - C), \quad B = (2C\hat{x}_0^\pm + D)(\pm F - C), \quad J = (2C\hat{x}_0^\pm + D) - 2U(\hat{x}_0^\pm)(\pm F + C).$$

(X, Y) :

$$x = \hat{x}_0^\pm + \frac{X - B/A}{(X - B/A)^2 + Y^2}, \quad y = \frac{Y}{(X - B/A)^2 + Y^2}$$

:

$$AX^2 - JY^2 = (\xi_0^\pm)^2/4.$$

$$(4a) \quad (4c), \quad f \in \mathbb{R} :$$

$$X = \frac{\cosh f}{s_1}, \quad Y = \frac{\sinh f}{s_2}, \quad s_1 \equiv \sqrt{4A/(\xi_0^\pm)^2} \in \mathbb{R}, \quad s_2 \equiv \sqrt{4J/(\xi_0^\pm)^2} \in \mathbb{R}.$$

$$f \quad \tau. \quad (x, y) \mapsto (X, Y),$$

$$f' = \frac{\pm |U(\hat{x}_0^\pm)| \xi_0^\pm}{\beta(x(f), y(f))} = \pm |U(\hat{x}_0^\pm)| \xi_0^\pm \frac{P(\cosh f)}{S(\cosh f)}$$

$$S(s) = -U_0 P(s) \mp F - (2C\hat{x}_0^\pm + D) \left(\frac{s}{s_1} - \frac{B}{A} \right),$$

$$P = \frac{s_2^2 + s_1^2}{s_2^2} \left(\left(\frac{s}{s_1} \right)^2 - s \frac{2B}{As_1} \frac{s_2^2}{s_1^2 + s_2^2} + \left(\frac{B^2}{A^2} - 1/s_2^2 \right) \frac{s_2^2}{s_1^2 + s_2^2} \right).$$

, f :

$$\tau - \tau_0 = \frac{\pm 1}{|U(\hat{x}_0^\pm)| \xi_0^\pm} \int_{f_0}^f \frac{S(\cosh f)}{P(\cosh f)} df = \quad (14)$$

$$\frac{\pm 1}{\xi_0^\pm} (f - f_0) \pm \frac{s_1 s_2^2 (2C\hat{x}_0^\pm + D)}{|U(\hat{x}_0^\pm, 0)| \xi_0^\pm (s_1^2 + s_2^2) (X_2 - X_1)} \left(\left(X_1 - \left(\frac{B}{A} \mp \frac{F}{2C\hat{x}_0^\pm + D} \right) \right) I(f, X_1 s_1) - \right.$$

$$- \left(X_2 - \left(\frac{B}{A} \mp \frac{F}{2C\hat{x}_0^\pm + D} \right) \right) I(f, X_2 s_1) \Bigg),$$

$$I(f, a) \equiv \int_{f_0}^f \frac{dx}{\cosh x - a} = \frac{2}{\sqrt{1-a^2}} \arctan \sqrt{\frac{1+a}{1-a}} \tanh \frac{x}{2} \Bigg|_{f_0}^f.$$

$X_1, X_2 \in P$:

$$X_{1,2} = \frac{Bs_2^2}{A(s_1^2 + s_2^2)} \pm \frac{2s_2^2(\pm F - C)}{(s_1^2 + s_2^2)\xi_0^\pm} \sqrt{\frac{-C}{U_0}}$$

, $M_{0,+} \cap M_{0,-} = \emptyset$ (3.3) $(x_0^\pm, 0)$, $\quad$:

$$x(\tau, \tau_0) = x_0^\pm + \frac{\frac{\cosh(f(\tau - \tau_0))}{s_1} - \frac{B}{A}}{P(\cosh(f(\tau - \tau_0)))}, \quad y(\tau, \tau_0) = \frac{\sinh(f(\tau - \tau_0))}{s_2 P(\cosh(f(\tau - \tau_0)))}, \quad \tau_0 \in \mathbb{R}.$$

$$f(\tau) \text{ (14).}$$

3.4.

$$, \quad (\hat{x}_0^+, 0, \lambda_+(\hat{x}_0^+, 0), 0), (\hat{x}_0^-, 0, \lambda_-(\hat{x}_0^-, 0), 0) \quad (\cdot, \cdot), \quad \cdot \quad (\pm \cdot):$$

$$\pm \frac{\sqrt{q \pm \sqrt{p}}}{\sqrt{2}},$$

(M_0)

$$p = \left(4\varepsilon^2(F^2 - C^2) \mp |\alpha| |U(\hat{x}_0^\pm, 0)| \right)^2 + 8\alpha\varepsilon^2(F \mp C)(2C\hat{x}_0^\pm + D),$$

$$q = -4\varepsilon^2(F^2 - C^2) \mp |\alpha| |U(\hat{x}_0^\pm, 0)|.$$

1 :

- $(\hat{x}_0^-, 0, \lambda_-(\hat{x}_0^-, 0), 0) \in M_{0,-} \text{ --- } - \text{ } (\quad, \quad, \quad)$
- $(\hat{x}_0^+, 0, \lambda_+(\hat{x}_0^+, 0), 0) \in M_{0,+} \text{ --- } - \text{ } (\pm i\omega, \pm \xi).$

$$\omega = \frac{\sqrt{\sqrt{p}-q}}{\sqrt{2}} = \sqrt{|\alpha U(\hat{x}_0^+, 0)|} + \mathcal{O}(\varepsilon^2),$$

$$\xi = \frac{\sqrt{\sqrt{p}+q}}{\sqrt{2}} = \mathcal{O}(\varepsilon),$$

4.

4.1.

$$, \quad \cdot \quad, \quad [?].$$

:

$$\begin{cases} \frac{dX}{dt} = JD_X H_1(X, I) + \varepsilon g^X(X, I), \\ \frac{dI}{dt} = \varepsilon g^I(X, I), \end{cases} \quad (15)$$

$$X \in \mathbb{R}^2, I \in \mathbb{R}^m, D_X, D_I \text{ --- }, J \text{ --- }.$$

$$V \subset M_0 \text{ --- } M_0 \text{ (15), } V_\varepsilon \text{ --- } \varepsilon \text{---} V, \quad \cdot \quad \Gamma = W^s(V) \cap W^u(V) \setminus V.$$

$$W^s(V_\varepsilon) \cap W^u(V_\varepsilon) \cap p \in \Gamma : \quad$$

$$d^I(p, t_0, \varepsilon) = \varepsilon \frac{M^I(p, t_0)}{\|D_X H(X_0^I, I)\|} + O(\varepsilon^2), \quad (16)$$

$$M^I - \dots, \quad :$$

$$M^I(p, t_0) = \int_{-\infty}^{\infty} \left(\langle D_X H_1, g^X \rangle + \langle D_I H_1, g^I \rangle \right) (X_0^I(t - t_0, I), I) dt - \\ - \langle D_I H_1(v(I), I), \int_{-\infty}^{\infty} g^I(X_0^I(t - t_0, I), I) dt \rangle.$$

$$X = v(I) - V, \langle \cdot, \cdot \rangle - \mathbb{R}^2. \\ W^s(V_\varepsilon) \cap W^u(V_\varepsilon) \cap \varepsilon > 0 : \quad$$

$$M^I = 0, \quad \nabla M^I \neq 0. \quad (17)$$

$$, \quad , \quad \varepsilon \rightarrow d^I \quad .$$

4.2.

$$, \quad M_0 \quad M_{0,+}, M_{0,-} \quad M_{0,b}^\pm, \quad M_{0,+} \quad , \quad M_{0,-} \quad . \\ , \quad M_{0,-} \quad . \\ , \quad (12) \quad M_{0,-} \quad (x(t), y(t), 0, \lambda_-(x(t), y(t)), \Lambda = 0) \quad xy \quad . \\ (5) \quad (15) \quad :$$

$$X = \begin{pmatrix} \Lambda \\ z \end{pmatrix}, \quad I = \begin{pmatrix} x \\ y \end{pmatrix}, \quad z = \lambda - \lambda_-(x, y).$$

$$(5) \quad (15):$$

$$D_X = \begin{pmatrix} \frac{\partial}{\partial \Lambda} \\ \frac{\partial}{\partial z} \end{pmatrix}, \quad D_I = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \end{pmatrix}, \quad H_1 = \alpha \frac{\Lambda^2}{2} + \beta(x, y)(1 - \cos z),$$

$$g^I = \begin{pmatrix} -2Fy + \frac{\partial \beta}{\partial y} \cos z + \beta(x, y) \frac{\partial}{\partial y} \arctan\left(\frac{V}{U}\right) \sin z \\ 2Fx + G - \frac{\partial \beta}{\partial x} \cos z - \beta(x, y) \frac{\partial}{\partial x} \arctan\left(\frac{V}{U}\right) \sin z \end{pmatrix}, \quad g^X = \frac{-1}{\varepsilon} \begin{pmatrix} 0 \\ \frac{d}{dt} \arctan\left(\frac{V}{U}\right) \end{pmatrix}.$$

4.3.

$$V = M_{0,-} \quad :$$

$$M^I(x, y) = \frac{4}{\alpha \beta^2} \left(U(x, y) \left((2Fx + G) \frac{\partial U}{\partial y} - 2Fy \frac{\partial U}{\partial x} \right) + V(x, y) \left((2Fx + G) \frac{\partial V}{\partial y} - 2Fy \frac{\partial V}{\partial x} \right) \right). \quad (18)$$

:

$$0 = y \left((2Cx + D)^2 + C^2 y^2 + (2Cx + D) \frac{F(D^2 - 4CE)}{GC - FD} + \frac{2(D^2 - 4CE)}{4} \right), \quad (19)$$

$$y = 0 \quad (. \ 2).$$

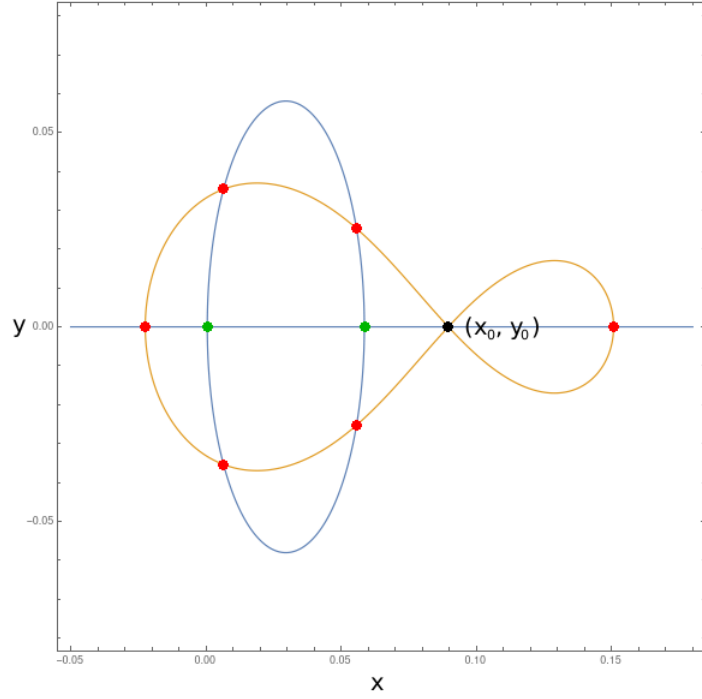
$$, \quad , \quad .$$

1. $\left\{ (x(\tau), y(\tau), 0, \lambda_-(x(\tau), y(\tau))), \tau \in \mathbb{R} \right\} \in M_{0,-} - (12) \quad M_{0,-}, \quad D_0 - \delta - M_{0,-}, \quad (x_b^\pm, y_b). \quad D_0$
- $M_{0,-} \quad \varepsilon > 0 \quad D_\varepsilon \quad (5), \quad .$
2. $\varepsilon > 0 \quad (x_0^-, y_0^-, 0, \lambda_-(x_0^-, y_0^-)) \quad (12). \quad , \quad \varepsilon > 0 \quad , \quad , \quad .$

: $D_0 \quad M_{0,-} \quad \varepsilon > 0 \quad D_\varepsilon, \quad \varepsilon \quad D_0.$, , .

$$(x_0^-, y_0^-, 0, \lambda_-(x_0^-, y_0^-))$$

(3.3), (5). , , . (. . 2), . ■
, , $M_{0,-}$,



. 2. () (). - M^I

4.4.

- 0, (5) :

$$\begin{aligned} \dot{\Lambda} &= U \sin \lambda - V \cos \lambda, & \dot{\lambda} &= \alpha \Lambda, \\ \dot{x} &= -\varepsilon \left(2Fy + \frac{\partial U}{\partial y} \cos \lambda + \frac{\partial V}{\partial y} \sin \lambda \right), & \dot{y} &= \varepsilon \left(2Fx + (2Fx_0^- + G) + \frac{\partial U}{\partial x} \cos \lambda + \frac{\partial V}{\partial x} \sin \lambda \right), \end{aligned} \quad (20)$$

$$\begin{aligned} U(x, y) &= C(x^2 - y^2) + Dx + (E + Dx_0^- + C(x_0^-)^2 + 2Cx_0^-) \equiv C(x^2 - y^2) + Dx + U_0, \\ V(x, y) &= 2C(x + x_0^-)y + Dy \end{aligned}$$

ε , : ε .

4.4.1.

$$\begin{aligned} \Lambda(t) &= \Lambda_0(t) + \varepsilon \Lambda_1(t) + O(\varepsilon^2), & \lambda(t) &= \lambda_0(t) + \varepsilon \lambda_1(t) + O(\varepsilon^2), \\ x(t) &= x_0(t) + \varepsilon x_1(t) + O(\varepsilon^2), & y(t) &= y_0(t) + \varepsilon y_1(t) + O(\varepsilon^2). \end{aligned} \quad (21)$$

$$(21) \quad (20) \quad \varepsilon, \quad .$$

:

$$\dot{\Lambda}_0 = U(\lambda_0) \sin \lambda_0 - V(\lambda_0) \cos \lambda_0, \quad \dot{\lambda}_0 = \alpha \Lambda_0, \quad (22)$$

$$\dot{x}_0 = 0, \quad \dot{y}_0 = 0$$

$$\Lambda_0 = \lambda_0, \quad x_0, y_0, \quad \Lambda_0(t) \rightarrow 0, \lambda_0(t) \rightarrow \pi \pm \pi \quad t \rightarrow \pm \infty, \quad , \quad . \quad x_0 = y_0 :$$

$$x_0(t) \equiv y_0(t) \equiv 0.$$

$$x_0, y_0$$

$$\frac{d}{dt} \begin{pmatrix} \Lambda_1 \\ \lambda_1 \end{pmatrix} = \begin{pmatrix} 0 & U \cos \lambda_0 + V \sin \lambda_0 \\ \alpha & 0 \end{pmatrix} \begin{pmatrix} \Lambda_1 \\ \lambda_1 \end{pmatrix} + \begin{pmatrix} G_{1\Lambda} \\ 0 \end{pmatrix}, \quad (23)$$

$$G_{1\Lambda}(t) = x_1(t)V_y \sin \lambda_0(t) - y_1(t)V_y \cos \lambda_0(t).$$

$$\Lambda_1, \lambda_1, \quad (\Lambda_0(t), \lambda_0(t)).$$

$$(x_1, y_1) :$$

$$\dot{x}_1 = -V_y \sin \lambda_0(t),$$

$$(24)$$

$$\dot{y}_1 = (2Fx_0^- + G) + V_y \cos \lambda_0(t).$$

$$(24) \quad x_1(t) = y_1(t) \quad .$$

$$(21) \quad |\varepsilon t e^{\omega|t|}| < 1, \quad t - \quad . \quad , \quad , \quad .$$

4.4.2.

$$(20) \quad :$$

$$\frac{d}{dt} \begin{pmatrix} \Lambda \\ \lambda \\ x \\ y \end{pmatrix} = A \begin{pmatrix} \Lambda \\ \lambda \\ x \\ y \end{pmatrix} + \mathcal{G}(\Lambda, \lambda, x, y), \quad (25)$$

$$A - (20) \quad , \quad \mathcal{G} - \quad .$$

$$, \quad t \rightarrow +\infty \quad () \quad t \rightarrow -\infty \quad () . \quad , \quad .$$

$$R, \quad A \quad A \quad \text{diag}(\omega, -\omega, \xi, -\xi):$$

$$R = \begin{pmatrix} a & -a & b & -b \\ c & c & d & d \\ -g & g & -e & e \\ 1 & 1 & 1 & 1 \end{pmatrix}, \quad (26)$$

$$a, b, c, d, e - \quad , \quad .$$

:

$$X_s(t) = R \begin{pmatrix} 0 \\ c_1 e^{-\omega t} \\ 0 \\ c_2 e^{-\xi t} \end{pmatrix} + R \sum_{k+j \geq 2}^{\infty} T_{k,j}^s c_1^k c_2^j e^{-(k\omega + j\xi)t},$$

$$X_u(t) = R \begin{pmatrix} c_3 e^{\omega t} \\ 0 \\ c_4 e^{\xi t} \\ 0 \end{pmatrix} + R \sum_{k+j \geq 2}^{\infty} T_{k,j}^u c_3^k c_4^j e^{-(k\omega + j\xi)t}$$

$$(c_1, c_2) = W^s(0), (c_3, c_4) = W^u(0), T_{k,j}^{s,u} = \dots$$

:

$$|c_1|e^{-\omega t} + \varepsilon|c_2|e^{-\xi t} < 1,$$

$$|c_1|e^{\omega t} + \varepsilon|c_2|e^{\xi t} < 1.$$

..

4.4.3.

$$(\Lambda, \lambda, x, y, t), \quad (21) \quad () \quad () = \dots, \quad t, \quad, \quad x, y \quad.$$

$$, \quad t^* > 0. \quad t = t^*, \quad (21) \quad \varepsilon, \quad t = -t^* \quad.$$

$$(\Lambda_1, \lambda_1) :$$

$$\begin{pmatrix} \Lambda_1^s(t) \\ \lambda_1^s(t) \end{pmatrix} = \mathbf{\Omega}(t, t^*) \begin{pmatrix} \Delta_{s,\Lambda} \\ \Delta_{s,\lambda} \end{pmatrix} - \int_t^{t^*} \mathbf{\Omega}(t, \tau) \begin{pmatrix} G_{1,\Lambda} \\ 0 \end{pmatrix} d\tau, \quad (27)$$

$$\begin{pmatrix} \Lambda_1^u(t) \\ \lambda_1^u(t) \end{pmatrix} = \mathbf{\Omega}(t, -t^*) \begin{pmatrix} \Delta_{u,\Lambda} \\ \Delta_{u,\lambda} \end{pmatrix} + \int_{-t^*}^t \mathbf{\Omega}(t, \tau) \begin{pmatrix} G_{1,\Lambda} \\ 0 \end{pmatrix} d\tau, \quad (28)$$

$$\mathbf{\Omega}(t, \tau) \quad (23).$$

:

$$\begin{pmatrix} x_1^s(t) \\ y_1^s(t) \end{pmatrix} = \begin{pmatrix} \Delta_{s,x} \\ \Delta_{s,y} \end{pmatrix} - \int_t^{t^*} \begin{pmatrix} -V_y \sin \lambda_0(\tau) \\ (2Fx_0^- + G) + V_y \cos \lambda_0(\tau) \end{pmatrix} d\tau, \quad (29)$$

$$\begin{pmatrix} x_1^u(t) \\ y_1^u(t) \end{pmatrix} = \begin{pmatrix} \Delta_{u,x} \\ \Delta_{u,y} \end{pmatrix} + \int_{-t^*}^t \begin{pmatrix} -V_y \sin \lambda_0(\tau) \\ (2Fx_0^- + G) + V_y \cos \lambda_0(\tau) \end{pmatrix} d\tau. \quad (30)$$

$$\Delta_{s,\rho}, \Delta_{u,\rho}, \quad \rho \in \{\Lambda, \lambda, x, y\} \quad t^* () \quad -t^* () :$$

$$\begin{pmatrix} \Lambda_0(t^*) \\ \lambda_0(t^*) \\ 0 \\ 0 \end{pmatrix} + \varepsilon \begin{pmatrix} \Delta_{s,\Lambda} \\ \Delta_{s,\lambda} \\ \Delta_{s,x} \\ \Delta_{s,y} \end{pmatrix} = c_1 e^{-\omega t^*} \begin{pmatrix} -a \\ c \\ g \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} -b \\ d \\ e \\ 1 \end{pmatrix} \quad (31)$$

$$\begin{pmatrix} \Lambda_0(-t^*) \\ \lambda_0(-t^*) \\ 0 \\ 0 \end{pmatrix} + \varepsilon \begin{pmatrix} \Delta_{u,\Lambda} \\ \Delta_{u,\lambda} \\ \Delta_{u,x} \\ \Delta_{u,y} \end{pmatrix} = c_3 e^{-\omega t^*} \begin{pmatrix} a \\ c \\ -g \\ 1 \end{pmatrix} + c_4 \begin{pmatrix} b \\ d \\ -e \\ 1 \end{pmatrix} \quad (32)$$

$$, \quad, \quad \varepsilon \rightarrow 0 \quad t \quad.$$

4.4.4. $W^u(0) \quad W^s(0)$

-. , :

$$\begin{pmatrix} \Lambda^u(t) \\ \lambda^u(t) \\ x^u(t) \\ y^u(t) \end{pmatrix} - \begin{pmatrix} \Lambda^s(t) \\ \lambda^s(t) \\ x^s(t) \\ y^s(t) \end{pmatrix} \quad (33)$$

$$(33) \quad c_1, c_2, c_3, c_4, t.$$

$$(33) \quad, \quad (27)-(30) \quad:$$

$$\begin{aligned} \varepsilon \int_{-t^*}^{t^*} \left(\begin{pmatrix} -V_y \sin \lambda_0(\tau) \\ (2Fx_0^- + G) + V_y \cos \lambda_0(\tau) \end{pmatrix} d\tau \right) &= e^{-\omega t^*} \begin{pmatrix} g(c_1 + c_3) \\ c_1 - c_3 \end{pmatrix} + \begin{pmatrix} e(c_2 + c_4) \\ c_2 - c_4 \end{pmatrix} = \\ &= e^{-\omega t^*} \begin{pmatrix} 0 & g \\ -1 & 0 \end{pmatrix} \begin{pmatrix} c_3 - c_1 \\ c_1 + c_3 \end{pmatrix} + \begin{pmatrix} 0 & e \\ -1 & 0 \end{pmatrix} \begin{pmatrix} c_2 - c_2 \\ c_2 + c_4 \end{pmatrix} \quad (34) \end{aligned}$$

$$\int_{-t*}^{t*} \mathbf{\Omega}(t, \tau) \begin{pmatrix} G^{1, \Lambda} \\ 0 \end{pmatrix} d\tau = \mathbf{\Omega}(t, t*) \begin{pmatrix} \Delta_{s, \Lambda} \\ \Delta_{s, \lambda} \end{pmatrix} - \mathbf{\Omega}(t, -t*) \begin{pmatrix} \Delta_{u, \Lambda} \\ \Delta_{u, \lambda} \end{pmatrix} \quad (35)$$

$$, \quad (23)$$

$$(\Lambda, \lambda, t) \mapsto (\Lambda, -\lambda, -t).$$

$$\vdots$$

$$\mathbf{\Omega}(-t, -s) = R \mathbf{\Omega}(t, s) R, \quad R = \text{diag}(1, -1).$$

$$\Delta_{s,u,\Lambda,\lambda} \quad :$$

$$\begin{aligned} \varepsilon \int_{-t_*}^{t_*} \mathbf{\Omega}(t, \tau) \begin{pmatrix} G_{1,\Lambda} \\ 0 \end{pmatrix} d\tau = e^{-\omega t_*} (-c_1 R \mathbf{\Omega}(0, -t_*) - c_3 \mathbf{\Omega}(0, t_*)) \begin{pmatrix} a \\ c \end{pmatrix} + \\ + (-c_2 R \mathbf{\Omega}(0, -t_*) - c_4 \mathbf{\Omega}(0, t_*)) \begin{pmatrix} b \\ f \end{pmatrix} + (-R \mathbf{\Omega}(0, -t_*) + \mathbf{\Omega}(0, t_*)) \begin{pmatrix} \Lambda_0(-t_*) \\ \lambda_0(-t_*) \end{pmatrix}. \quad (36) \end{aligned}$$

$$\mathbf{\Omega}(0, -t^*) \equiv \begin{pmatrix} \varphi_{11} & \varphi_{12} \\ \varphi_{21} & \varphi_{22} \end{pmatrix},$$

$$(36) :$$

$$\begin{aligned} \varepsilon \int_{-t*}^{t*} \mathbf{\Omega}(t, \tau) \begin{pmatrix} G_{1,\Lambda} \\ 0 \end{pmatrix} d\tau = e^{-\omega t*} \begin{pmatrix} a\varphi_{11} + c\varphi_{12} & 0 \\ 0 & a\varphi_{21} + c\varphi_{22} \end{pmatrix} \begin{pmatrix} c_3 - c_1 \\ c_1 + c_3 \end{pmatrix} + \\ \begin{pmatrix} b\varphi_{11} + d\varphi_{12} & 0 \\ 0 & b\varphi_{21} + d\varphi_{22} \end{pmatrix} \begin{pmatrix} c_2 - c_2 \\ c_2 + c_4 \end{pmatrix} + \\ (0 \quad 2\varphi_{21}\Lambda_0(-t*) + 2\varphi_{22}\lambda_0(-t*)). \quad (37) \end{aligned}$$

5.

$$(\hat{x}_0^+, 0, \lambda_+(\hat{x}_0^+, 0), 0) \in M_{0,+}, \quad \text{if } \lambda_+(\hat{x}_0^+, 0) \geq \varepsilon, \quad \text{if } \lambda_+(\hat{x}_0^+, 0) < \varepsilon.$$

5.1.

$$- \frac{1}{2} \frac{\partial}{\partial t} \left(\frac{1}{\rho} \frac{\partial \rho}{\partial t} \right) + \frac{1}{2} \frac{\partial}{\partial t} \left(\frac{1}{\rho} \frac{\partial \rho}{\partial t} \right) = 0,$$

$$\Lambda \rightarrow \Lambda, \quad \lambda \rightarrow \lambda - \lambda_+(\hat{x}_0, 0) = \lambda - \pi, \quad x \rightarrow x - \hat{x}_0, \quad y \rightarrow y.$$

$$\vdots$$

$$\begin{cases} \dot{\Lambda} = -U \sin \lambda + V \cos \lambda \equiv g^\Lambda(\Lambda, \lambda, x, y), \\ \dot{x} = -\varepsilon(2Fy - \frac{\partial U}{\partial y} \cos \lambda - \frac{\partial V}{\partial y} \sin \lambda) \equiv g^x(\Lambda, \lambda, x, y), \end{cases} \quad \begin{cases} \dot{\lambda} = \alpha \Lambda \equiv g^\lambda(\Lambda, \lambda, x, y), \\ \dot{y} = \varepsilon(2F(x + \hat{x}_0) + G - \frac{\partial U}{\partial x} \cos \lambda - \frac{\partial V}{\partial x} \sin \lambda) \equiv g^y(\Lambda, \lambda, x, y). \end{cases} \quad (38)$$

$$U(x, y) = U_0 + x(2C\hat{x}_0 + D) + C(x^2 - y^2), \quad V(x, y) = 2Cxy + y(2C\hat{x}_0 + D),$$

$$U_0 = U(\hat{x}_0^+, 0).$$

, (38) (??) :

$$H = \frac{\alpha\Lambda^2}{2} - U(x, y) \cos \lambda - V(x, y) \sin \lambda + F((x + \hat{x}_0)^2 + y^2) + Gx. \quad (39)$$

5.1.1.

$$(38), \quad W^{s,u}(0) \quad . \quad (38) \quad :$$

• $W^s(0)$:

$$\Lambda^s(+\infty) = 0, \quad \lambda^s(+\infty) = 0, \quad x^s(+\infty) = 0, \quad y^s(+\infty) = 0,$$

• $W^u(0)$:

$$\Lambda^u(-\infty) = 0, \quad \lambda^u(-\infty) = 0, \quad x^u(-\infty) = 0, \quad y^u(-\infty) = 0.$$

(38) :

$$\Lambda^{s,u}(\tau) = \sum_{k=0}^{\infty} \varepsilon^k \Lambda_k^{s,u}(\tau), \quad \lambda^{s,u}(\tau) = \sum_{k=0}^{\infty} \varepsilon^k \lambda_k^{s,u}(\tau), \quad x^{s,u}(\tau) = \sum_{k=0}^{\infty} \varepsilon^k x_k^{s,u}(\tau), \quad y^{s,u}(\tau) = \sum_{k=0}^{\infty} \varepsilon^k y_k^{s,u}(\tau), \quad (40)$$

, $\tau = \varepsilon t$. ε , , , 2 , $(\lambda_k, \Lambda_k), k \geq 1$:

$$\begin{cases} \Lambda_k(\tau) = \frac{1}{\alpha} \lambda'_{k-1}, \\ \lambda_k(\tau) = \frac{1}{\beta} \left(\alpha^{-1} \lambda''_{k-2} - G_k^\Lambda + x_k((2Cx_0 + 2C\hat{x}_0 + D) \sin \lambda_0 - 2Cy_0 \cos \lambda_0) + \right. \\ \quad \left. + y_k(-2Cy_0 \sin \lambda_0 - (2Cx_0 + 2C\hat{x}_0 + D) \cos \lambda_0) \right), \\ \lambda_{-1} = 0, \end{cases} \quad (41)$$

$(x_k, y_k), k \geq 1$ - :

$$\frac{d}{d\tau} \begin{pmatrix} x_k \\ y_k \end{pmatrix} = \begin{pmatrix} a(\tau) & b(\tau) \\ c(\tau) & -a(\tau) \end{pmatrix} \begin{pmatrix} x_k \\ y_k \end{pmatrix} + \begin{pmatrix} \tilde{G}_k^x(\tau) \\ \tilde{G}_k^y(\tau) \end{pmatrix} \equiv \mathcal{A}(\tau) \begin{pmatrix} x_k \\ y_k \end{pmatrix} + \begin{pmatrix} \tilde{G}_k^x(\tau) \\ \tilde{G}_k^y(\tau) \end{pmatrix}, \quad (42)$$

:

$$\tilde{G}_k^x = G_k^x - \beta^{-1}(\alpha^{-1} \lambda''_{k-2} - G_k^\Lambda)(-2Cy_0 \sin \lambda_0 - (2Cx_0 + 2C\hat{x}_0 + D) \cos \lambda_0),$$

$$\tilde{G}_k^y = G_k^y - \beta^{-1}(\alpha^{-1} \lambda''_{k-2} - G_k^\Lambda)((2Cx_0 + 2C\hat{x}_0 + D) \sin \lambda_0 + 2Cy_0 \cos \lambda_0).$$

a, b, c - $x_0(\tau), y_0(\tau), \lambda_0(\tau), \Lambda_0(\tau)$:

$$a(\tau) = \frac{(2D^2 - 4CE^2) \sin \lambda_0 \cos \lambda_0}{\beta(x_0, y_0)}, \quad (43)$$

$$b(\tau) = -2F - 2C \cos \lambda_0 - \frac{1}{\beta(x_0, y_0)} ((2Cy_0) \sin \lambda_0 + (2Cx_0 + 2C\hat{x}_0 + D) \cos \lambda_0)^2, \quad (44)$$

$$c(\tau) = 2F - 2C \cos \lambda_0 + \frac{1}{\beta(x_0, y_0)} ((2Cx_0 + 2C\hat{x}_0 + D) \sin \lambda_0 - (2Cy_0) \cos \lambda_0)^2. \quad (45)$$

$G_k^\rho, \rho \in \{\lambda, \Lambda, x, y\}$, (42), :

$$G_k^\rho = \sum_{j=2}^k \frac{\mathcal{L}_k^{(j)} g^\rho}{j!}, \quad k \geq 2,$$

$$g^\rho = \quad, \quad \rho \in \{\lambda, \Lambda, x, y\},$$

$$\mathcal{L}_k^{(j)} = \sum_{k_1+\dots+k_j=k} \mathfrak{D}_{k_1} \cdot \dots \cdot \mathfrak{D}_{k_j}, \quad \mathfrak{D}_k = \left(\begin{pmatrix} \lambda_k \\ \Lambda_k \\ x_k \\ y_k \end{pmatrix}, \begin{pmatrix} \frac{\partial}{\partial \lambda} \Big|_{\lambda_0, \Lambda_0, x_0, y_0} \\ \frac{\partial}{\partial \Lambda} \Big|_{\lambda_0, \Lambda_0, x_0, y_0} \\ \frac{\partial}{\partial x} \Big|_{\lambda_0, \Lambda_0, x_0, y_0} \\ \frac{\partial}{\partial y} \Big|_{\lambda_0, \Lambda_0, x_0, y_0} \end{pmatrix} \right),$$

$$(\cdot, \cdot) = \quad \mathbb{R}^4.$$

$$:$$

$$\mathcal{F}_s = \left\{ f: \mathbb{R} \rightarrow \mathbb{R} \mid \exists C > 0: \forall x \in \mathbb{R} \Rightarrow |f(x)| \leq C e^{-s|x|} \right\}, s > 0$$

$$\mathcal{F}_{-s} = \left\{ f: \mathbb{R} \rightarrow \mathbb{R} \mid \exists C > 0: \forall x \in \mathbb{R} \Rightarrow |f(x)| \geq C e^{s|x|} \right\}, s > 0$$

$$\lambda_j, \Lambda_j, x_j, y_j \in \mathcal{F}_s \quad \forall j: \quad 0 \leq j \leq k, \quad g^\rho \in C_b^\infty(\mathbb{R}^4), \quad \forall \rho \in \{\Lambda, \lambda, x, y\} \quad (\quad, \quad), \quad \quad:$$

$$G_{k+1}^\rho \in \mathcal{F}_{2s} \quad \forall \rho \in \{\Lambda, \lambda, x, y\}$$

$$, \quad:$$

$$G_k^\lambda = 0 \quad \forall k \geq 1.$$

$$\varepsilon = 0.$$

$$, \quad W^{s,u}(0), \quad:$$

$$\begin{cases} \Lambda_0(\tau) = 0, & \lambda_0(\tau) = \lambda_-(x_{sep}(\tau), y_{sep}(\tau)), \\ x_0(\tau) = x_{sep}(\tau) - \hat{x}_0, & y_0(\tau) = y_{sep}(\tau). \end{cases}$$

$$(x_1, y_1) \quad:$$

$$\frac{d}{d\tau} \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \mathcal{A}(\tau) \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}. \tag{46}$$

$$\quad \quad \quad (\quad). \quad \tilde{G}_1^x \quad \tilde{G}_1^y \quad - \quad .$$

$$(46) \quad:$$

$$x_1^{s,u}(\tau) = \frac{dx_0}{d\tau}, \quad y_1^{s,u}(\tau) = \frac{dy_0}{d\tau}, \quad \mathbf{u}_1(\tau) \equiv \begin{pmatrix} x_1(\tau) \\ y_1(\tau) \end{pmatrix} \tag{47}$$

$$(46) \quad (\quad) \quad \tilde{\mathbf{u}}_1(\tau) \equiv (\tilde{x}_1(\tau), \tilde{y}_1(\tau)). \dots \quad \mathcal{A} \quad 0, \quad, \quad -, \quad, \quad, \quad: \quad$$

$$W(\tau) = x_1(\tau)\tilde{y}_1(\tau) - y_1(\tau)\tilde{x}_1(\tau) = 1$$

$$(x_1, y_1) \quad (\tilde{x}_1, \tilde{y}_1).$$

$$\mathbf{u}_1(\tau) \quad \tilde{\mathbf{u}}_1(\tau) \quad. \quad \mathcal{A}(\tau) \quad |\tau|:$$

$$\mathcal{A}(\tau) = \begin{pmatrix} 0 & -2F - 2C + \frac{(2C\hat{x}_0 + D)^2}{U_0} \\ 2F - 2C & 0 \end{pmatrix} + \mathcal{O}(e^{-\xi_0^+|\tau|}) \equiv \begin{pmatrix} 0 & b_0 \\ c_0 & 0 \end{pmatrix} + \mathcal{O}(e^{-\xi_0^+|\tau|}), \quad \sqrt{b_0 c_0} = \xi_0^+ > 0$$

$$(48)$$

$$:$$

$$\mathbf{u}_1(\tau) = \text{const} \begin{pmatrix} b_0 \sqrt{c_0} \\ c_0 \sqrt{b_0} \end{pmatrix} e^{-\xi_0^+|\tau|} + \mathcal{O}(e^{-2\xi_0^+|\tau|}) \in \mathcal{F}_{\xi_0^+}, \quad \tilde{\mathbf{u}}_1(\tau) = \text{const} \begin{pmatrix} b_0 \sqrt{c_0} \\ -c_0 \sqrt{b_0} \end{pmatrix} e^{\xi_0^+|\tau|} + \mathcal{O}(1) \in \mathcal{F}_{-\xi_0^+}.$$

$$\begin{aligned} & , \quad \mathbf{u}_1(\tau) \quad W^s(0), \quad W^u(0), \quad s, u \quad . \\ & \mathcal{A}(\tau), \quad , \quad , \quad : \end{aligned}$$

$$\begin{cases} \mathbf{u}_k^u = \mathbf{u}_1(\tau) \int_{\tau_k^u}^{\tau} \left(\tilde{y}_1(\tau) \tilde{G}_k^x(\tau) - \tilde{x}_1(\tau) \tilde{G}_k^y(\tau) \right) d\tau + \tilde{\mathbf{u}}_1(\tau) \int_{-\infty}^{\tau} \left(x_1(\tau) \tilde{G}_k^y(\tau) - y_1(\tau) \tilde{G}_k^x(\tau) \right) d\tau, \\ \mathbf{u}_k^s = \mathbf{u}_1(\tau) \int_{\tau_k^s}^{\tau} \left(\tilde{y}_1(\tau) \tilde{G}_k^x(\tau) - \tilde{x}_1(\tau) \tilde{G}_k^y(\tau) \right) d\tau - \tilde{\mathbf{u}}_1(\tau) \int_{\tau}^{+\infty} \left(x_1(\tau) \tilde{G}_k^y(\tau) - y_1(\tau) \tilde{G}_k^x(\tau) \right) d\tau. \end{cases} \quad (49)$$

$$\begin{aligned} & , \quad (49) \quad \tau \rightarrow \pm\infty. , \quad (49) \quad \tau \quad . \\ & \mathbf{u}_1(\tau) \quad \tau \rightarrow \pm\infty, \quad \mathbf{u}_k^u \quad \mathbf{u}_k^s \quad \tau_k^u, \tau_k^s \in \mathbb{R}. \\ & \tilde{\mathbf{u}}_1(\tau) \quad \tau \rightarrow \pm\infty. \quad \mathbf{u}_k^u \quad \mathbf{u}_k^s \quad , \quad (49), \quad \tilde{\mathbf{u}}_1(\tau) \quad . \quad (x_1 \tilde{G}_k^y - y_1 \tilde{G}_k^x) \in \mathcal{F}_{3\xi_0^+}, \quad , \quad , \quad \tilde{\mathbf{u}}_1. \\ & (40), \quad (x_1, y_1) \quad (42) \quad (x_1(\tau), y_1(\tau)) = \text{const} \cdot \mathbf{u}_1, \quad \tau_k^s \quad . \\ & - \quad , \quad W^s(0) \quad . \quad (38), \quad W^s(0) \quad . \\ & W^u(0), \quad , \quad +\infty. \\ & : \end{aligned}$$

$$\begin{aligned} |\mathbf{u}_k^u(\tau) - \mathbf{u}_k^s(\tau)| &= \left| \mathbf{u}_1(\tau) \int_{\tau_k^u}^{\tau_k^s} \left(\tilde{y}_1(\tau) \tilde{G}_k^x(\tau) - \tilde{x}_1(\tau) \tilde{G}_k^y(\tau) \right) d\tau + \right. \\ & \quad \left. + \tilde{\mathbf{u}}_1(\tau) \int_{-\infty}^{+\infty} \left(x_1(\tau) \tilde{G}_k^y(\tau) - y_1(\tau) \tilde{G}_k^x(\tau) \right) d\tau \right| \quad (50) \end{aligned}$$

$$\begin{aligned} & \tau \quad . \quad W^s(0) \quad W^u(0) \quad \mathbf{u}_k^s \quad \mathbf{u}_k^u \quad \tau. \\ & , \quad |\mathbf{u}_k^u(\tau_1) - \mathbf{u}_k^s(\tau_2)| = 0 \quad \tau_1, \tau_2. , \quad , \quad \tau_1 = \tau_2 = \tau. \end{aligned}$$

$$1. \quad , \quad f(\tau) \quad , \quad f(\tau) \quad , \quad \tau.$$

$$\begin{aligned} & , \quad \lambda_j, \Lambda_j, x_j, y_j, \quad j \leq k-1 \quad \tau \quad . \quad (38) \quad , \quad G_k^x(\tau), \tilde{G}_k^x(\tau), G_k^\Lambda(\tau), G_k^y(\tau) \quad . \quad G_k^x(\tau), \tilde{G}_k^x(\tau), G_k^\Lambda(\tau) \quad , \\ & G_k^y(\tau), \tilde{G}_k^y(\tau) \quad G_k^x(\tau) \quad (\quad). \\ & (38) \quad G_2^x(\tau), \tilde{G}_2^x(\tau), G_2^\Lambda \quad - \quad , \quad G_2^y(\tau), \tilde{G}_2^y(\tau) \quad - \quad . \\ & , \quad , \quad : \end{aligned}$$

$$\int_{-\infty}^{+\infty} \left(x_1(\tau) \tilde{G}_2^y(\tau) - y_1(\tau) \tilde{G}_2^x(\tau) \right) d\tau = 0. \quad (51)$$

5.1.2.

$$\begin{aligned} & , \quad 0. \\ & , \quad : \end{aligned}$$

$$\mathbf{u}_1^0(\tau) \equiv \begin{pmatrix} x_1^0(\tau) \\ y_1^0(\tau) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

$$\begin{aligned} & \tilde{G}_2^\rho, \rho \in \{\Lambda, \lambda, x, y\} \quad . \\ & \tau_2^s = \tau_2^u = 0 \quad : \end{aligned}$$

$$\begin{cases} \mathbf{u}_2^u = \mathbf{u}_1(\tau) \int_0^{\tau} \left(\tilde{y}_1(\tau) \tilde{G}_2^x(\tau) - \tilde{x}_1(\tau) \tilde{G}_2^y(\tau) \right) d\tau + \tilde{\mathbf{u}}_1(\tau) \int_{-\infty}^{\tau} \left(x_1(\tau) \tilde{G}_2^y(\tau) - y_1(\tau) \tilde{G}_2^x(\tau) \right) d\tau, \\ \mathbf{u}_2^s = \mathbf{u}_1(\tau) \int_0^{\tau} \left(\tilde{y}_1(\tau) \tilde{G}_2^x(\tau) - \tilde{x}_1(\tau) \tilde{G}_2^y(\tau) \right) d\tau - \tilde{\mathbf{u}}_1(\tau) \int_{\tau}^{+\infty} \left(x_1(\tau) \tilde{G}_2^y(\tau) - y_1(\tau) \tilde{G}_2^x(\tau) \right) d\tau, \end{cases}$$

(51)

$$\mathbf{u}_2^s(\tau) = \mathbf{u}_2^u(\tau) \equiv \mathbf{u}_2^0(\tau).$$

$$\tilde{G}_k^\rho, \rho \in \{\Lambda, \lambda, x, y\} \quad 3 \quad :$$

$$\mathbf{u}_3^0(\tau) \equiv \begin{pmatrix} x_3^0(\tau) \\ y_3^0(\tau) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad (\lambda_3^0(\tau), \Lambda_3^0(\tau)) = (0, 0).$$

$$k \geq 0 :$$

$$G_{2k+1}^\rho = 0, \rho \in \{\Lambda, \lambda, x, y\}, \quad G_{2k+2}^x, y_{2k+2} - , \quad G_{2k+2}^y, x_{2k+2} - .$$

$$(\lambda_{2k+1}^0, \Lambda_{2k+1}^0, x_{2k+1}^0, y_{2k+1}^0) = (0, 0, 0, 0).$$

$$\begin{cases} \lambda^0(\tau) = (\lambda_0(\tau) - \pi) + \sum_{k=1}^{+\infty} \varepsilon^{2k} \lambda_{2k}^0(\tau), & \Lambda^0(\tau) = \frac{1}{\alpha} \sum_{k=1}^{+\infty} \varepsilon^{2k} (\lambda_{2k-1}^0)'(\tau), \\ x^0(\tau) = x_0(\tau) + \sum_{k=1}^{+\infty} \varepsilon^{2k} x_{2k}^0(\tau), & y^0(\tau) = y_0(\tau) + \sum_{k=1}^{+\infty} \varepsilon^{2k} y_{2k}^0(\tau), \end{cases}$$

$$\begin{cases} x_{2k}^0(\tau) = x_1(\tau) \int_0^\tau \left(\tilde{y}_1(\tau) \tilde{G}_{2k}^x(\tau) - \tilde{x}_1(\tau) \tilde{G}_{2k}^y(\tau) \right) d\tau + \tilde{x}_1(\tau) \int_{-\infty}^\tau \left(x_1(\tau) \tilde{G}_{2k}^y(\tau) - y_1(\tau) \tilde{G}_{2k}^x(\tau) \right) d\tau, \\ y_{2k}^0(\tau) = y_1(\tau) \int_0^\tau \left(\tilde{y}_1(\tau) \tilde{G}_{2k}^x(\tau) - \tilde{x}_1(\tau) \tilde{G}_{2k}^y(\tau) \right) d\tau + \tilde{y}_1(\tau) \int_{-\infty}^\tau \left(x_1(\tau) \tilde{G}_{2k}^y(\tau) - y_1(\tau) \tilde{G}_{2k}^x(\tau) \right) d\tau, \end{cases}$$

$$\tilde{G}_{2k}^x = \frac{1}{\alpha\beta} (\lambda_{2k-2}^0)''(\tau) (-2Cy_0 \sin \lambda_0 - (2Cx_0 + 2C\hat{x}_0 + D) \cos \lambda_0),$$

$$\tilde{G}_{2k}^y = \frac{1}{\alpha\beta} (\lambda_{2k-2}^0)''(\tau) (-(2Cx_0 + 2C\hat{x}_0 + D) \sin \lambda_0 + 2Cy_0 \cos \lambda_0).$$

2. (38):

- $(\Lambda^s(\tau), \lambda^s(\tau), x^s(\tau), y^s(\tau)) \in W^s(0),$
- $(\Lambda^u(\tau), \lambda^u(\tau), x^u(\tau), y^u(\tau)) \in W^u(0),$

$$\varepsilon \quad (\Lambda^0(\tau), \lambda^0(\tau), x^0(\tau), y^0(\tau)). \quad c > 0 :$$

$$(\Lambda^s(\tau), \lambda^s(\tau), x^s(\tau), y^s(\tau)) = (\Lambda^0(\tau), \lambda^0(\tau), x^0(\tau), y^0(\tau)) + \mathcal{O}(e^{-\frac{c}{\varepsilon}}),$$

$$(\Lambda^u(\tau), \lambda^u(\tau), x^u(\tau), y^u(\tau)) = (\Lambda^0(\tau), \lambda^0(\tau), x^0(\tau), y^0(\tau)) + \mathcal{O}(e^{-\frac{c}{\varepsilon}}).$$

$$\int_0^\tau \left(\tilde{y}_1(\tau) \tilde{G}_{2k}^x(\tau) - \tilde{x}_1(\tau) \tilde{G}_{2k}^y(\tau) \right) d\tau.$$

$$\mathcal{F}_{\xi_0^+}, \quad k \in \mathbb{N}.$$

$$\int_{-\infty}^{+\infty} \left(x_1(\tau) \tilde{G}_{2k}^y(\tau) - y_1(\tau) \tilde{G}_{2k}^x(\tau) \right) d\tau = 0, \quad k \in \mathbb{N}.$$

$$\mathcal{F}_{3\xi_0^+},$$

$$\int_{-\infty}^\tau \left(x_1(\tau) \tilde{G}_{2k}^y(\tau) - y_1(\tau) \tilde{G}_{2k}^x(\tau) \right) d\tau$$

$$\mathcal{F}_{2\xi_0^+ \cdot}, \quad \tau \rightarrow \pm\infty \quad \tilde{u}_1.$$

$$(\Lambda^0(\tau), \lambda^0(\tau), x^0(\tau), y^0(\tau))$$

$$\tau \rightarrow \pm\infty.$$

$$(g^\Lambda, g^\lambda, g^x, g^y) \quad (38) \quad - \quad (\Lambda, \lambda, x, y) \quad \varepsilon > 0. \quad , \quad (g^\Lambda, g^\lambda, g^x, g^y) \quad . \quad , \quad (38) \quad \varepsilon \left(., , [?] \right).$$

$$(\Lambda^s, \lambda^s, x^s, y^s) \in W^s(0), \quad (\Lambda^u, \lambda^u, x^u, y^u) \in W^u(0),$$

$$\varepsilon:$$

$$(\Lambda^0(\tau), \lambda^0(\tau), x^0(\tau), y^0(\tau)).$$

$$W^s(0) \cap W^u(0) \quad , \quad (\Lambda^0, \lambda^0, x^0, y^0) \quad .$$

$$(\Lambda^s - \Lambda^u, \lambda^s - \lambda^u, x^s - x^u, y^s - y^u)$$

$$\varepsilon.$$

$$, \quad , \quad , \quad (., , \text{????}[?]).$$

$$, \quad c > 0 \quad ,$$

$$(\Lambda^s(\tau), \lambda^s(\tau), x^s(\tau), y^s(\tau)) = (\Lambda^0(\tau), \lambda^0(\tau), x^0(\tau), y^0(\tau)) + \mathcal{O}(e^{-c/\varepsilon}) \quad ,$$

$$(\Lambda^u(\tau), \lambda^u(\tau), x^u(\tau), y^u(\tau)) = (\Lambda^0(\tau), \lambda^0(\tau), x^0(\tau), y^0(\tau)) + \mathcal{O}(e^{-c/\varepsilon}) \quad .$$

■

$$1. \quad - \quad (38) \quad .$$

$$5.2.$$

$$[?], \quad - \quad . \quad [?] \quad [?, ?]. \quad .$$

$$5.2.1.$$

$$(38) \quad . \quad :$$

$$A = \begin{pmatrix} 0 & -U_0 & 0 & (2C\hat{x}_0 + D) \\ \alpha & 0 & 0 & 0 \\ 0 & \varepsilon(2C\hat{x}_0 + D) & 0 & -\varepsilon(2F + 2C) \\ 0 & 0 & \varepsilon(2F - 2C) & 0 \end{pmatrix}.$$

$$, \quad A \quad \pm i\omega, \pm\xi \quad (z, \eta, a, b)^T = R^{-1}(\Lambda, \lambda, x, y)^T, \quad R - \quad . \quad :$$

$$\frac{d}{dt}(z, \eta, a, b)^T = \text{diag}(i\omega, -i\omega, \xi, -\xi)(z, \eta, a, b)^T + \mathcal{G}(z, \eta, a, b), \quad (52)$$

$$\mathcal{G} \quad .$$

$$\varphi(t), \psi(t), \quad , \quad [?] \quad . \quad :$$

$$\varphi(t) = \varphi_0 e^{i\theta(\varphi_0, \psi_0)t}, \quad \psi(t) = \varphi_0 e^{-i\theta(\varphi_0, \psi_0)t}.$$

$$\theta(\varphi_0, \psi_0) \quad \varphi_0, \psi_0:$$

$$\theta(\varphi_0, \psi_0) = \omega + \sum_{k=1}^{+\infty} \theta_k \cdot (\varphi_0 \psi_0)^k.$$

$$\begin{cases} z(\varphi, \psi) = \varphi + \sum_{k,j=1}^{\infty} \{z\}_{k,j} \varphi^k \psi^j, & \eta(\varphi, \psi) = \psi + \sum_{k,j=1}^{\infty} \{\eta\}_{k,j} \varphi^k \psi^j, \\ a(\varphi, \psi) = \sum_{k,j=1}^{\infty} \{a\}_{k,j} \varphi^k \psi^j, & b(\varphi, \psi) = \sum_{k,j=1}^{\infty} \{b\}_{k,j} \varphi^k \psi^j. \end{cases} \quad (53)$$

$$\begin{aligned} & , \quad z = (\varphi\psi)^k \varphi, \quad \eta = (\varphi\psi)^k \psi. \quad \varphi, \psi \text{ [?]}. \\ (52) \quad & : \end{aligned}$$

$$\begin{cases} ((p-q)i\omega - i\omega) \{z\}_{p,q} + \sum_{r=1}^{+\infty} (p-q)i\theta_r \{z\}_{p-r,q-r} = \{\mathcal{G}_z(z, \eta, a, b)\}_{p,q}, \\ ((p-q)i\omega + i\omega) \{\eta\}_{p,q} + \sum_{r=1}^{+\infty} (p-q)i\theta_r \{\eta\}_{p-r,q-r} = \{\mathcal{G}_\eta(z, \eta, a, b)\}_{p,q}, \\ ((p-q)i\omega - \xi) \{a\}_{p,q} + \sum_{r=1}^{+\infty} (p-q)i\theta_r \{a\}_{p-r,q-r} = \{\mathcal{G}_a(z, \eta, a, b)\}_{p,q}, \\ ((p-q)i\omega + \xi) \{b\}_{p,q} + \sum_{r=1}^{+\infty} (p-q)i\theta_r \{b\}_{p-r,q-r} = \{\mathcal{G}_b(z, \eta, a, b)\}_{p,q}. \end{cases} \quad (54)$$

$$\begin{aligned} & \{\mathcal{G}_\rho(z, \eta, a, b)\}_{p,q}, \rho \in \{z, \eta, a, b\} \quad \varphi^p \psi^q \quad \mathcal{G}. \\ & p = q + 1, p + q > 1 \quad z, \quad \{z\}_{p,q} = 0. \quad q = p + 1 \quad \eta, \quad \{\eta\}_{p,q} = 0 \quad (\varphi\psi)^k \varphi \quad z \quad (\varphi\psi)^k \psi \quad \eta). \\ & \{z\}_{1,0} = \{\eta\}_{0,1} = 1. < \end{aligned}$$

$$\theta_p = \{\mathcal{G}_z\}_{p,p-1} = -\{\mathcal{G}_z\}_{p-1,p} \quad p > 1$$

$$(\Lambda, \lambda, x, y), \quad : \varphi_0 = \psi_0 \equiv h.$$

5.2.2.

.

$$2. \quad , \quad f(x), \quad - \quad 0,$$

$$g(s) = \sum_{n=0}^{+\infty} g_n s^n, \quad s \geq 0, \quad g_n \geq 0 \quad \forall n \geq 0,$$

:

$$f(x) \quad :$$

$$f(x) = \sum_{n=0}^{+\infty} f_n x^n.$$

$$\exists C > 0 : |f_n| \leq C \cdot g_n \quad \forall n \geq 0.$$

$$f \prec g.$$

$\varkappa$:

$$\begin{aligned} \varkappa &= \max \left(\sup_{p,q \in \mathbb{R}} \left(\left| \frac{1}{(p-q)i\omega - \xi} \right| \right), \sup_{p,q \in \mathbb{R}} \left(\left| \frac{p-q}{(p-q)i\omega - \xi} \right| \right), \sup_{p,q \in \mathbb{R}} \left(\left| \frac{1}{(p-q)i\omega - i\omega} \right| \right), \sup_{p,q \in \mathbb{R}} \left(\left| \frac{p-q}{(p-q)i\omega - i\omega} \right| \right) \right) = \\ &= \max \left(\frac{1}{|\xi|}, \frac{1}{|\omega|} \right) = \frac{1}{|\xi|} = O \left(\frac{1}{\varepsilon} \right) \quad \varepsilon. \end{aligned}$$

$$, \quad (54), \quad :$$

$$|\{\rho\}_{p,q}| \leq \varkappa |\{\mathcal{G}_\rho(z, \eta, a, b)\}_{p,q}| + \varkappa \sum_{r=1}^{+\infty} |(p-q)\theta_r \{\rho\}_{p-r,q-r}|, \quad \rho \in \{z, \eta, a, b\} \quad (55)$$

$$[?] \quad , \quad \mathcal{G}$$

$$\mathcal{G} \prec \frac{1}{1-s}, \quad s \equiv |z| + |\eta| + |a| + |b|.$$

$\mathcal{G}$.

$$(\Lambda, \lambda, x, y) \prec (z, \eta, a, b) \iff R, \quad s:$$

$$s = |z| + |\eta| + \varepsilon(|a| + |b|).$$

$R :$

$$\Lambda \prec s, \quad \lambda \prec s, \quad x \prec \varepsilon s, \quad y \prec \varepsilon s. \quad (56)$$

$$(56) \quad , \quad 0, \quad (38):$$

$$\begin{cases} -U \sin \lambda + V \cos \lambda \prec (\varepsilon s + \varepsilon^2 s^2) e^s, & \alpha \Lambda \prec s, \\ -\varepsilon(2Fy - \frac{\partial U}{\partial y} \cos \lambda - \frac{\partial V}{\partial y} \sin \lambda) \prec \varepsilon(\varepsilon s + \varepsilon^2 s^2) e^s, & \varepsilon(2F(x + \hat{x}_0) + G - \frac{\partial U}{\partial x} \cos \lambda - \frac{\partial V}{\partial x} \sin \lambda) \prec \varepsilon(\varepsilon s + \varepsilon^2 s^2) e^s. \end{cases}$$

$\mathcal{G} :$

$$\begin{cases} \mathcal{G}_{z,\eta} \prec \varepsilon((s + \varepsilon s^2) e^s - s), \\ \mathcal{G}_{a,b} \prec ((s + \varepsilon s^2) e^s - s). \end{cases} \quad (57)$$

$$, \quad -s \quad (s + \varepsilon s^2) e^s \quad (s + \varepsilon s^2) e^s - s \quad .$$

$h, \quad \varphi, \psi, :$

$$||h|| \equiv \sum_{p,q} |\{h\}_{p,q}| \varphi^p \psi^q.$$

$$(55) \quad \varepsilon \ a \ b, \quad \varphi^p \psi^q \quad . \quad :$$

$$(\varphi + \psi) ||\theta - \omega|| + S \prec \varkappa(||\mathcal{G}_z|| + ||\mathcal{G}_\eta|| + \varepsilon(||\mathcal{G}_a|| + ||\mathcal{G}_b||) + ||\theta - \omega|| S), \quad (58)$$

$$S = ||z - \varphi|| + ||\eta - \psi|| + \varepsilon(||a|| + ||b||).$$

$$\varphi = \psi > 0 \quad . \quad :$$

$$2\varphi ||\theta - \omega|| + S \prec \varkappa(\varepsilon(s + \varepsilon s^2) e^s - \varepsilon s + ||\theta - \omega|| S),$$

$$s \prec 2\varphi + S.$$

$$Y = 2||\theta - \omega|| + \frac{1}{\varphi} S, \quad :$$

$$Y \prec \frac{\varkappa}{4} (Y^2 + \varepsilon((2 + Y) + \varepsilon\varphi(2 + Y)^2) e^{\varphi(2+Y)} - (2 + Y)). \quad (59)$$

$$(59), \quad , \quad R = \mathcal{O}(\varepsilon), \quad \varphi.$$

5.3.

$$, \quad , \quad .$$

:

$$\Gamma_h(t) = (\Lambda^*(h, t), \lambda^*(h, t), x^*(h, t), y^*(h, t)). \quad (60)$$

$$W^s(\Gamma_h) \cap W^u(\Gamma_h) \neq \emptyset, \quad , \quad :$$

$$\begin{cases} \Lambda^{s,u}(h, t) = \Lambda^*(h, t) + z_\Lambda^{s,u}(h, t), & \lambda^{s,u}(h, t) = \lambda^*(h, t) + z_\lambda^{s,u}(h, t), \\ x^{s,u}(h, t) = x^*(h, t) + z_x^{s,u}(h, t), & y^{s,u}(h, t) = y^*(h, t) + z_y^{s,u}(h, t), \end{cases} \quad (61)$$

$$z_\Lambda^{s,u}, z_\lambda^{s,u}, z_x^{s,u}, z_y^{s,u} \quad :$$

$$\bullet \quad W^s(\Gamma_h):$$

$$z_\Lambda^s(t) \rightarrow 0, \quad z_\lambda^s(t) \rightarrow 0, \quad z_x^s(t) \rightarrow 0, \quad z_y^s(t) \rightarrow 0 \quad t \rightarrow +\infty;$$

• $W^u(\Gamma_h)$:

$$z_\Lambda^u(t) \rightarrow 0, z_\lambda^u(t) \rightarrow 0, z_x^u(t) \rightarrow 0, z_y^u(t) \rightarrow 0 \quad t \rightarrow -\infty.$$

$$\begin{aligned} \Theta(h) &= h, \quad \tilde{t} = \frac{\theta(h)}{\omega} t = (1 + \mathcal{O}(h^2))t, \quad h \rightarrow h, \quad \omega, \quad \hat{H}(\mathbf{z}, \tilde{t}) = \mathbf{z}, \quad \cdot \\ \omega = \mathcal{O}(1) & \quad \hat{H} \quad \xi = \mathcal{O}(\varepsilon), \quad (\langle \cdot \rangle), \quad : \end{aligned}$$

$$\begin{aligned} \langle \hat{H} \rangle &= H(\mathbf{z}) - h^2 \theta_1 H(\mathbf{z}) + \alpha \langle \Lambda^* \rangle z_\Lambda + 2F \langle x^* \rangle z_x - \\ &\quad - \langle \cos \lambda^* - 1 \rangle \left(U(z_x, z_y) \cos z_\lambda + V(z_x, z_y) \sin z_\lambda \right) - \\ &\quad - \cos z_\lambda \left((2C \hat{x}_0 + D) \langle y^* \sin \lambda^* \rangle_0 + z_x 2C (\langle x^* \cos \lambda^* \rangle + \langle y^* \sin \lambda^* \rangle) + \langle (U(x^*, y^*) - U_0) \cos \lambda^* \rangle \right) - \\ &\quad - \sin z_\lambda \left(z_y 2C (\langle x^* \cos \lambda^* \rangle + \langle y^* \sin \lambda^* \rangle) \right) + \mathcal{O}(h^4). \quad (62) \end{aligned}$$

$h, \langle \hat{H} \rangle$

$$(\hat{\Lambda}_0^*(h), \hat{\lambda}_0^*(h), \hat{x}_0^*(h), \hat{y}_0^*(h)), \quad (63)$$

$\mathcal{O}(h^2)$.

$$(63) \quad 0, \quad \mathcal{O}(h^4), \quad : \quad [?, ?], \quad (\quad), \quad h, \quad \cdot$$

$$\begin{aligned} \hat{H}_0 &= \left[\frac{\alpha \omega}{2\theta(h)} \right]_0 z_\Lambda^2 - \hat{U}(z_x, z_y) \cos z_\lambda - \hat{V}(z_x, z_y) \sin z_\lambda + \left[\frac{F\omega}{\theta(h)} \right]_0 ((z_x + \hat{x}_0 + [\hat{x}_0^*(h)]_0)^2 + z_y^2) + \\ &\quad + \left[\frac{\omega}{\theta(h)} \right]_0 (G + [2F \langle x^* \rangle_0]_0 + 2F \hat{x}_0^*(h)) z_x, \quad (64) \end{aligned}$$

$\hat{U}, \hat{V}, U, V, \mathcal{O}(h^2), [\cdot]_0, \mathcal{O}(h^4).$

$$, \quad h \quad (39) \quad (64) \quad (\quad) \quad \mathcal{O}(h^2)) \quad 1, \quad (64) \quad , \quad 0 - (64) \quad , \quad \cdot$$

5.3.1.

$$(60) \quad \mathcal{O}(\varepsilon) \quad (.. |h| \leq R(\varepsilon) = \mathcal{O}(\varepsilon)), \quad \mu \equiv h/\varepsilon:$$

$$|\mu| \leq \frac{R(\varepsilon)}{\varepsilon} = \mathcal{O}(1)$$

$$(64) \quad H(\mathbf{z}):$$

$$\hat{H}_0(\mathbf{z}) = H(\mathbf{z}) + \varepsilon^2 \mu^2 H_1(\mathbf{z}), \quad (65)$$

:

$$\begin{aligned} H_1(\mathbf{z}) &\equiv \left[-\theta_1 H_{old}(\mathbf{z}) + 2F z_x \underbrace{(-2\varepsilon \xi) \{a\}_{1,1}}_{\mathcal{O}(1)} - \underbrace{\frac{1}{4} \left(\frac{2(\xi^2 - \alpha U_0)}{(2C \hat{x}_0 + D)} \right)^2}_{\mathcal{O}(1)} \left(U(z_x, z_y) \cos z_\lambda + V(z_x, z_y) \sin z_\lambda \right) - \right. \\ &\quad \left. - \cos z_\lambda (2C z_x + (2C \hat{x}_0 + D)) \underbrace{(-2\varepsilon \xi) \{a\}_{1,1}}_{\mathcal{O}(1)} - \sin z_\lambda 2C z_y \underbrace{(-2\varepsilon \xi) \{a\}_{1,1}}_{\mathcal{O}(1)} \right]_0. \end{aligned}$$

$$(65) \quad (40):$$

$$z_\Lambda^{s,u}(\tau, \mu) = \sum_{k=0}^{\infty} \varepsilon^k z_{\Lambda,k}^{s,u}(\tau, \mu), \quad z_\lambda^{s,u}(\tau, \mu) = \sum_{k=0}^{\infty} \varepsilon^k z_{\lambda,k}^{s,u}(\tau, \mu), \quad z_x^{s,u}(\tau, \mu) = \sum_{k=0}^{\infty} \varepsilon^k z_{x,k}^{s,u}(\tau, \mu), \quad z_y^{s,u}(\tau, \mu) = \sum_{k=0}^{\infty} \varepsilon^k z_{y,k}^{s,u}(\tau, \mu), \quad (66)$$

:

- $W^s(0)$:

$$z_\Lambda^s(+\infty) = 0, \quad z_\lambda^s(+\infty) = 0, \quad z_x^s(+\infty) = 0, \quad z_y^s(+\infty) = 0,$$

- $W^u(0)$:

$$z_\Lambda^u(-\infty) = 0, \quad z_\lambda^u(-\infty) = 0, \quad z_x^u(-\infty) = 0, \quad z_y^u(-\infty) = 0.$$

(66) (65):

$$\begin{aligned} \frac{dz_\Lambda}{dt} &= -U \sin z_\lambda + V \cos z_\lambda - \varepsilon^2 \mu^2 \frac{\partial H_1}{\partial z_\lambda} + \mathcal{O}(\varepsilon^4), & \frac{dz_x}{dt} &= -\varepsilon \left(2F z_y - \frac{\partial U}{\partial z_y} \cos z_\lambda - \frac{\partial V}{\partial z_y} \sin z_\lambda + \varepsilon^2 \mu^2 \frac{\partial H_1}{\partial z_y} + \mathcal{O}(\varepsilon^4) \right), \\ \frac{dz_\lambda}{dt} &= \alpha z_\Lambda + \varepsilon^2 \mu^2 \frac{\partial H_1}{\partial z_\Lambda} + \mathcal{O}(\varepsilon^4), & \frac{dz_y}{dt} &= \varepsilon \left(2F(z_x + \hat{x}_0) + G - \frac{\partial U}{\partial z_x} \cos z_\lambda - \frac{\partial V}{\partial z_x} \sin z_\lambda + \varepsilon^2 \mu^2 \frac{\partial H_1}{\partial z_x} + \mathcal{O}(\varepsilon^4) \right). \end{aligned} \quad (67)$$

ε .

.

$$z_{\Lambda,0}(\tau) = 0, \quad z_{\lambda,0}(\tau) = \lambda_-(x_{sep}(\tau), y_{sep}(\tau)), \quad z_{x,0}(\tau) = x_{sep}(\tau) - \hat{x}_0, \quad z_{y,0}(\tau) = y_{sep}(\tau).$$

$$(z_{\Lambda,1}, z_{\lambda,1}, z_{x,1}, z_{y,1}) \quad (z_{\Lambda,1}, z_{\lambda,1}):$$

$$\begin{cases} z_{\Lambda,1}(\tau) = z'_{\lambda,0}, \\ z_{\lambda,1}(\tau) = \frac{1}{\beta} \left(z_{x,1} \left((2C z_{x,0} + 2C \hat{x}_0 + D) \sin z_{\lambda,0} - 2C z_{y,0} \cos z_{\lambda,0} \right) + \right. \\ \left. + z_{y,1} \left(-2C z_{y,0} \sin z_{\lambda,0} - (2C z_{x,0} + 2C \hat{x}_0 + D) \cos z_{\lambda,0} \right) \right), \end{cases}$$

$$(z_{x,1}, z_{y,1}):$$

$$\frac{d}{d\tau} \begin{pmatrix} z_{x,1} \\ z_{y,1} \end{pmatrix} = \mathcal{A}(\tau) \begin{pmatrix} z_{x,1} \\ z_{y,1} \end{pmatrix}$$

$$\mathcal{A}(\tau) - \mathbf{u}_1(\tau), \tilde{\mathbf{u}}_1(\tau).$$

$$W^s(\Gamma_h) \quad W^u(\Gamma_h), \quad . \quad :$$

$$(z_{x,1}(\tau), z_{y,1}(\tau)) \equiv (0, 0)$$

$$, \quad \mu, \quad \mu \quad .$$

$$(z_{x,2}, z_{y,2}) \quad , \quad \mu:$$

$$\frac{d}{d\tau} \begin{pmatrix} z_{x,2} \\ z_{y,2} \end{pmatrix} = \mathcal{A}(\tau) \begin{pmatrix} z_{x,2} \\ z_{y,2} \end{pmatrix} + \begin{pmatrix} \tilde{G}_2^x(\tau) \\ \tilde{G}_2^y(\tau) \end{pmatrix} + \mu^2 \begin{pmatrix} \mathfrak{G}_2^x(\tau) \\ \mathfrak{G}_2^y(\tau) \end{pmatrix}, \quad (68)$$

:

$$\begin{aligned} \mathfrak{G}_2^x &= \frac{\partial H_1}{\partial z_y} \Big|_{z_{\Lambda,0}, z_{\lambda,0}, z_{x,0}, z_{y,0}} - \\ &\quad - \beta^{-1} \left(\frac{\partial H_1}{\partial z_\lambda} \Big|_{z_{\Lambda,0}, z_{\lambda,0}, z_{x,0}, z_{y,0}} \right) \left(-2C z_{y,0} \sin z_{\lambda,0} - (2C z_{x,0} + 2C \hat{x}_0 + D) \cos z_{\lambda,0} \right), \end{aligned}$$

$$\begin{aligned} \mathfrak{G}_2^y &= \frac{\partial H_1}{\partial z_x} \Big|_{z_{\Lambda,0}, z_{\lambda,0}, z_{x,0}, z_{y,0}} - \\ &\quad - \beta^{-1} \left(\frac{\partial H_1}{\partial z_\lambda} \Big|_{z_{\Lambda,0}, z_{\lambda,0}, z_{x,0}, z_{y,0}} \right) \left(- (2C z_{x,0} + 2C \hat{x}_0 + D) \sin z_{\lambda,0} + 2C z_{y,0} \cos z_{\lambda,0} \right), \end{aligned}$$

$$\tilde{G}_2^x, \tilde{G}_2^y - \quad (42), \quad \mathbf{z}_0, \mathbf{z}_1.$$

$$\begin{cases} z_{\Lambda,2}(\tau) = \frac{1}{\alpha} z'_{\lambda,1}(\tau), \\ z_{\lambda,2}(\tau) = \frac{1}{\beta} \left(\alpha^{-1} z_{\lambda,0} + \frac{\partial H_1}{\partial z_{\lambda}} \Big|_{z_{\Lambda,0}, z_{\lambda,0}, z_{x,0}, z_{y,0}} + z_{x,2}((2C z_{x,0} + 2C \hat{x}_0 + D) \sin z_{\lambda,0} - 2C z_{y,0} \cos z_{\lambda,0}) + \right. \\ \left. + z_{y,2}(-2C z_{y,0} \sin z_{\lambda,0} - (2C z_{x,0} + 2C \hat{x}_0 + D) \cos z_{\lambda,0}) \right). \end{cases}$$

$$\begin{aligned} & \mathfrak{G}_2^x(\tau) - , \quad \mathfrak{G}_2^y(\tau) - . \\ & , \quad (68) \quad (42) \quad , \quad (68) \quad : \end{aligned}$$

$$\begin{cases} z_{x,2}(\tau, \mu) = z_{x,2}^0(\tau) + \hat{z}_{x,2}(\tau, \mu), \\ z_{y,2}(\tau, \mu) = z_{y,2}^0(\tau) + \hat{z}_{y,2}(\tau, \mu), \end{cases}$$

$$(z_{x,2}^0, z_{y,2}^0) - \quad (42), \quad (\hat{z}_{x,2}, \hat{z}_{y,2}) - :$$

$$\frac{d}{d\tau} \begin{pmatrix} \hat{z}_{x,2} \\ \hat{z}_{y,2} \end{pmatrix} = \mathcal{A}(\tau) \begin{pmatrix} \hat{z}_{x,2} \\ \hat{z}_{y,2} \end{pmatrix} + \mu^2 \begin{pmatrix} \mathfrak{G}_2^x(\tau) \\ \mathfrak{G}_2^y(\tau) \end{pmatrix},$$

$$\mathbf{u}_2 = (z_{x,2}, z_{y,2}), \quad \mathbf{u}_2^0 = (z_{x,2}^0, z_{y,2}^0), \quad \hat{\mathbf{u}}_2 = (\hat{z}_{x,2}, \hat{z}_{y,2}). \quad \hat{\mathbf{u}}_2^{s,u} \quad (s \quad u \quad):$$

$$\begin{cases} \hat{\mathbf{u}}_2^u = \mu^2 \mathbf{u}_1(\tau) \int_{\tau^u}^{\tau} (\tilde{y}_1(\tau) \mathfrak{G}_2^x(\tau) - \tilde{x}_1(\tau) \mathfrak{G}_2^y(\tau)) d\tau + \mu^2 \tilde{\mathbf{u}}_1(\tau) \int_{-\infty}^{\tau} (x_1(\tau) \mathfrak{G}_2^y(\tau) - y_1(\tau) \mathfrak{G}_2^x(\tau)) d\tau, \\ \hat{\mathbf{u}}_2^s = \mu^2 \mathbf{u}_1(\tau) \int_{\tau^s}^{\tau} (\tilde{y}_1(\tau) \mathfrak{G}_2^x(\tau) - \tilde{x}_1(\tau) \mathfrak{G}_2^y(\tau)) d\tau - \mu^2 \tilde{\mathbf{u}}_1(\tau) \int_{\tau}^{+\infty} (x_1(\tau) \mathfrak{G}_2^y(\tau) - y_1(\tau) \mathfrak{G}_2^x(\tau)) d\tau. \end{cases}$$

$$, \quad W^u(\Gamma_{h_1}) \quad W^s(\Gamma_{h_2}),$$

$$\mathbf{u}_2^u(\tau_1, \mu_1) - \mathbf{u}_2^s(\tau_2, \mu_2) = (\mathbf{u}_2^{0,u}(\tau_1) - \mathbf{u}_2^{0,s}(\tau_2)) + (\hat{\mathbf{u}}_2^u(\tau_1, \mu_1) - \hat{\mathbf{u}}_2^s(\tau_2, \mu_2)) = 0, \quad (69)$$

$$\mu_1^2, \mu_2^2$$

$$\tau^u = \tau_1, \quad \tau^s = \tau_2.$$

$$\hat{\mathbf{u}}_2^u, \hat{\mathbf{u}}_2^s \quad (69) \quad :$$

$$B(\tau_1, \tau_2) \cdot \begin{pmatrix} \mu_1^2 \\ \mu_2^2 \end{pmatrix} = \begin{pmatrix} f_1(\tau_1, \tau_2) \\ f_2(\tau_1, \tau_2) \end{pmatrix}, \quad (70)$$

$$B_{1,1} = \tilde{x}_1(\tau_1) \int_{-\infty}^{\tau_1} (x_1 \mathfrak{G}_2^y - y_1 \mathfrak{G}_2^x) d\tau, \quad B_{1,2} = \tilde{x}_1(\tau_2) \int_{\tau_2}^{+\infty} (x_1 \mathfrak{G}_2^y - y_1 \mathfrak{G}_2^x) d\tau,$$

$$B_{2,1} = \tilde{y}_1(\tau_1) \int_{-\infty}^{\tau_1} (x_1 \mathfrak{G}_2^y - y_1 \mathfrak{G}_2^x) d\tau, \quad B_{2,2} = \tilde{y}_1(\tau_2) \int_{\tau_2}^{+\infty} (x_1 \mathfrak{G}_2^y - y_1 \mathfrak{G}_2^x) d\tau.$$

$$f_1(\tau_1, \tau_2) = -x_2^{0,u}(\tau_1) + x_2^{0,s}(\tau_2) = -\tilde{x}_1(\tau_1) \int_{-\infty}^{\tau_1} (x_1 \tilde{G}_2^y - y_1 \tilde{G}_2^x) d\tau - \tilde{x}_1(\tau_2) \int_{\tau_2}^{+\infty} (x_1 \tilde{G}_2^y - y_1 \tilde{G}_2^x) d\tau,$$

$$f_2(\tau_1, \tau_2) = -y_2^{0,u}(\tau_1) + y_2^{0,s}(\tau_2) = -\tilde{y}_1(\tau_1) \int_{-\infty}^{\tau_1} (x_1 \tilde{G}_2^y - y_1 \tilde{G}_2^x) d\tau - \tilde{y}_1(\tau_2) \int_{\tau_2}^{+\infty} (x_1 \tilde{G}_2^y - y_1 \tilde{G}_2^x) d\tau.$$

$$, \quad (70) \quad \mu_1^2, \mu_2^2, \quad \tau_1, \tau_2. \\ \det B(\tau_1, \tau_2) \neq 0.$$

$$\det B(\tau_1, \tau_2) = \int_{-\infty}^{\tau_1} (x_1 \mathfrak{G}_2^y - y_1 \mathfrak{G}_2^x) d\tau \int_{\tau_2}^{+\infty} (x_1 \mathfrak{G}_2^y - y_1 \mathfrak{G}_2^x) d\tau (\tilde{x}_1(\tau_1) \tilde{y}_1(\tau_2) - \tilde{x}_1(\tau_2) \tilde{y}_1(\tau_1))$$

$$, \quad \mu_1, \mu_2 \in \mathbb{R}, \quad (70) \quad : \mu_1^2 > 0, \mu_2^2 > 0. \quad :$$

$$\frac{B_{1,1}f_2 - B_{2,1}f_1}{\det B} > 0, \quad \frac{B_{2,2}f_1 - B_{1,2}f_2}{\det B} > 0, \quad (71)$$

$$\frac{B_{1,1}f_2 - B_{2,1}f_1}{\det B} = -\frac{\int_{\tau_2}^{+\infty} (x_1 \tilde{G}_2^y - y_1 \tilde{G}_2^x) d\tau}{\int_{\tau_2}^{+\infty} (x_1 \mathfrak{G}_2^y - y_1 \mathfrak{G}_2^x) d\tau} > 0, \quad \frac{B_{2,2}f_1 - B_{1,2}f_2}{\det B} = -\frac{\int_{\tau_1}^{+\infty} (x_1 \tilde{G}_2^y - y_1 \tilde{G}_2^x) d\tau}{\int_{\tau_1}^{+\infty} (x_1 \mathfrak{G}_2^y - y_1 \mathfrak{G}_2^x) d\tau} > 0.$$

$$1. : \quad \frac{\int_{\tau_2}^{+\infty} (x_1 \tilde{G}_2^y - y_1 \tilde{G}_2^x) d\tau}{\int_{\tau_2}^{+\infty} (x_1 \mathfrak{G}_2^y - y_1 \mathfrak{G}_2^x) d\tau} < 0, \quad \frac{\int_{\tau_1}^{+\infty} (x_1 \tilde{G}_2^y - y_1 \tilde{G}_2^x) d\tau}{\int_{\tau_1}^{+\infty} (x_1 \mathfrak{G}_2^y - y_1 \mathfrak{G}_2^x) d\tau} < 0, \quad (72)$$

$$\tilde{y}_1(\tau_1) \tilde{x}_1(\tau_2) - \tilde{x}_1(\tau_1) \tilde{y}_1(\tau_2) \neq 0,$$

$$\tau_1 \neq \tau_2. \\ (70) \quad , \quad .$$

$$, \det B \quad , \quad (70) \quad . \\ , \quad . \blacksquare$$

$$2. \quad 1 \quad \tau_1 \neq \tau_2, \quad (38) \quad , \quad \Gamma_{h_1} \quad \Gamma_{h_2},$$

$$h_1 = \varepsilon \sqrt{-\frac{\int_{\tau_1}^{+\infty} (x_1 \tilde{G}_2^y - y_1 \tilde{G}_2^x) d\tau}{\int_{\tau_1}^{+\infty} (x_1 \mathfrak{G}_2^y - y_1 \mathfrak{G}_2^x) d\tau}}, \\ h_2 = \varepsilon \sqrt{-\frac{\int_{\tau_2}^{+\infty} (x_1 \tilde{G}_2^y - y_1 \tilde{G}_2^x) d\tau}{\int_{\tau_2}^{+\infty} (x_1 \mathfrak{G}_2^y - y_1 \mathfrak{G}_2^x) d\tau}}.$$

$$1 \quad h_1, h_2, \quad W^u(0) \quad (65), \quad h = h_1, \quad W^s(0) \quad (65), \quad h = h_2, \quad . \quad (??) \quad W^{s,u}(\Gamma_h) \quad \mathcal{O}(h^4). \quad , \quad (38) \\ \hat{h}_1, \hat{h}_2, \quad$$

$$\hat{h}_k = h_k + \mathcal{O}(\varepsilon^4)$$

$$W^u(\Gamma_{\hat{h}_1}) \quad W^s(\Gamma_{\hat{h}_2}). \quad . \quad \blacksquare \\ (72). \quad , \quad \tau_1, \tau_2. \quad$$

$$\tau_1 \quad \tau_2 \quad . \quad 11$$

$$\frac{\int_{\tau}^{+\infty} (x_1 \tilde{G}_2^y - y_1 \tilde{G}_2^x) d\tau}{\int_{\tau}^{+\infty} (x_1 \mathfrak{G}_2^y - y_1 \mathfrak{G}_2^x) d\tau} < 0$$

$$\tau.$$

$$[?] \quad 2 \quad .$$

6.

7.

7.1. 1

$$1) \quad g^\rho(\lambda, \Lambda, x, y) \quad (40). \quad :$$

$$g^\rho(\lambda, \Lambda, x, y) = g^\rho(\lambda_0, \Lambda_0, x_0, y_0) + \sum_{j=1}^{+\infty} \frac{\mathcal{T}^j g^\rho}{j!},$$

$\mathcal{T}$ - :

$$\begin{aligned} \mathcal{T} = & \left(\sum_{k=1}^{+\infty} \lambda_k \varepsilon^k \right) \frac{\partial}{\partial \lambda} \Big|_{\lambda_0, \Lambda_0, x_0, y_0} + \left(\sum_{k=1}^{+\infty} \Lambda_k \varepsilon^k \right) \frac{\partial}{\partial \Lambda} \Big|_{\lambda_0, \Lambda_0, x_0, y_0} + \\ & + \left(\sum_{k=1}^{+\infty} x_k \varepsilon^k \right) \frac{\partial}{\partial x} \Big|_{\lambda_0, \Lambda_0, x_0, y_0} + \left(\sum_{k=1}^{+\infty} y_k \varepsilon^k \right) \frac{\partial}{\partial y} \Big|_{\lambda_0, \Lambda_0, x_0, y_0}. \end{aligned}$$

$$\mathfrak{D}_k = \left(\begin{pmatrix} \lambda_k \\ \Lambda_k \\ x_k \\ y_k \end{pmatrix}, \begin{pmatrix} \frac{\partial}{\partial \lambda} \Big|_{\lambda_0, \Lambda_0, x_0, y_0} \\ \frac{\partial}{\partial \Lambda} \Big|_{\lambda_0, \Lambda_0, x_0, y_0} \\ \frac{\partial}{\partial x} \Big|_{\lambda_0, \Lambda_0, x_0, y_0} \\ \frac{\partial}{\partial y} \Big|_{\lambda_0, \Lambda_0, x_0, y_0} \end{pmatrix} \right),$$

:

$$\mathcal{T} = \sum_{k=1}^{+\infty} \varepsilon^k \mathfrak{D}_k.$$

$\mathcal{T}^j$, , :

$$\mathcal{T}^j = \left(\sum_{k=1}^{+\infty} \varepsilon^k \mathfrak{D}_k \right)^j = \sum_{k=j}^{+\infty} \varepsilon^k \mathcal{L}_k^{(j)},$$

:

$$\mathcal{L}_k^{(j)} = \sum_{k_1 + \dots + k_j = k} \mathfrak{D}_{k_1} \cdot \dots \cdot \mathfrak{D}_{k_j}.$$

ε :

$$g^\rho(\lambda, \Lambda, x, y) = g^\rho(\lambda_0, \Lambda_0, x_0, y_0) + \sum_{k=1}^{+\infty} \varepsilon^k \left(\sum_{j=1}^k \frac{\mathcal{L}_k^{(j)} g^\rho}{j!} \right).$$

2 , $\mathcal{L}_k^{(1)} = \mathfrak{D}_k$:

$$\begin{aligned}
g^\rho(\lambda, \Lambda, x, y) &= g^\rho(\lambda_0, \Lambda_0, x_0, y_0) + \varepsilon \mathfrak{D}_1 g^\rho + \sum_{k=2}^{+\infty} \varepsilon^k \left(\mathcal{L}_k^{(1)} g^\rho + \sum_{j=2}^k \frac{\mathcal{L}_k^{(j)} g^\rho}{j!} \right) = \\
&= g^\rho(\lambda_0, \Lambda_0, x_0, y_0) + \varepsilon \mathfrak{D}_1 g^\rho + \sum_{k=2}^{+\infty} \varepsilon^k \left(\mathfrak{D}_k g^\rho + \underbrace{\sum_{j=2}^k \frac{\mathcal{L}_k^{(j)} g^\rho}{j!}}_{G_k^\rho} \right)
\end{aligned}$$

- $\mathfrak{D}_k g^\rho - (\lambda_k, \Lambda_k, x_k, y_k) \quad ;$
- $G_k^\rho - (\lambda_j, \Lambda_j, x_j, y_j), \quad j \leq k-1 \quad (42).$

$$G_k^\rho \quad k \geq 2, \quad k=1 \quad G_1^\rho = 0. \quad :$$

$$G_k^\rho = \sum_{j=2}^k \frac{\mathcal{L}_k^{(j)} g^\rho}{j!}, \quad k \geq 2, \quad (73)$$

$$G_1^\rho \equiv 0,$$

$$g^\rho(\lambda, \Lambda, x, y) = g^\rho(\lambda_0, \Lambda_0, x_0, y_0) + \sum_{k=1}^{\infty} \varepsilon^k \mathfrak{D}_k g^\rho + \sum_{k=2}^{+\infty} \varepsilon^k G_k^\rho.$$

$$2) \quad g^\rho, \rho \in \{\lambda, \Lambda, x, y\} \quad , \quad (73) \quad j=2, \quad G_{k+1}^\rho \quad 2 \quad (\leq k) \quad \mathfrak{D}_i \mathfrak{D}_j g^\rho, i+j \leq k. \quad \leq k \quad \mathcal{F}_s, \quad , \quad ,$$

$$3) \quad g^\lambda = \alpha \Lambda \quad , \quad 0. \quad G_k^\lambda = 0 \quad \forall k \geq 1. \quad \blacksquare$$

7.2.

$$1) \quad 4 \quad - \quad \tau = \varepsilon t:$$

$$\begin{cases} \varepsilon \frac{d\Lambda}{d\tau} = g^\Lambda(\Lambda, \lambda, x, y), & \varepsilon \frac{d\lambda}{d\tau} = g^\lambda(\Lambda, \lambda, x, y), \\ \frac{dx}{d\tau} = g^x(\Lambda, \lambda, x, y), & \frac{dy}{d\tau} = g^y(\Lambda, \lambda, x, y). \end{cases} \quad (74)$$

$$\mathbf{X} = (\Lambda, \lambda, x, y) \quad : \quad \mathbf{X} = \sum_{k=0}^{\infty} \varepsilon^k \mathbf{X}_k. \quad \varepsilon \quad :$$

$$\begin{cases} 0 = g^\Lambda(\mathbf{X}_0), & 0 = g^\lambda(\mathbf{X}_0), \\ \frac{dx_0}{d\tau} = g^x(\mathbf{X}_0), & \frac{dy_0}{d\tau} = g^y(\mathbf{X}_0). \end{cases} \quad (75)$$

$$\begin{cases} \frac{d\Lambda_0}{d\tau} = (\nabla g^\Lambda(\mathbf{X}_0), \mathbf{X}_1), & \frac{d\lambda_0}{d\tau} = (\nabla g^\lambda(\mathbf{X}_0), \mathbf{X}_1), \\ \frac{dx_1}{d\tau} = (\nabla g^x(\mathbf{X}_0), \mathbf{X}_1), & \frac{dy_1}{d\tau} = (\nabla g^y(\mathbf{X}_0), \mathbf{X}_1). \end{cases} \quad (76)$$

$$, \quad (76) \quad (\Lambda_1, \lambda_1), \quad :$$

$$\begin{pmatrix} (\nabla g^\Lambda)_\Lambda & (\nabla g^\Lambda)_\lambda \\ (\nabla g^\lambda)_\Lambda & (\nabla g^\lambda)_\lambda \end{pmatrix} \begin{pmatrix} \Lambda_1 \\ \lambda_1 \end{pmatrix} = A \begin{pmatrix} \Lambda_1 \\ \lambda_1 \end{pmatrix} = \begin{pmatrix} \Lambda'_0 - (\nabla g^\Lambda)_x x_1 - (\nabla g^\Lambda)_y y_1 \\ \lambda'_0 - (\nabla g^\lambda)_x x_1 - (\nabla g^\lambda)_y y_1 \end{pmatrix}, \quad A \equiv \begin{pmatrix} (\nabla g^\Lambda)_\Lambda & (\nabla g^\Lambda)_\lambda \\ (\nabla g^\lambda)_\Lambda & (\nabla g^\lambda)_\lambda \end{pmatrix}.$$

:

$$\begin{pmatrix} \Lambda_1 \\ \lambda_1 \end{pmatrix} = M \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + \mathbf{C},$$

$$M = \frac{1}{\det A} \begin{pmatrix} -(\nabla g^\lambda)_\lambda (\nabla g^\Lambda)_x + (\nabla g^\Lambda)_\lambda (\nabla g^\lambda)_x & -(\nabla g^\lambda)_\lambda (\nabla g^\Lambda)_y + (\nabla g^\Lambda)_\lambda (\nabla g^\lambda)_y \\ (\nabla g^\lambda)_\Lambda (\nabla g^\Lambda)_x - (\nabla g^\Lambda)_\Lambda (\nabla g^\lambda)_x & (\nabla g^\lambda)_\Lambda (\nabla g^\Lambda)_y - (\nabla g^\Lambda)_\Lambda (\nabla g^\lambda)_y \end{pmatrix},$$

$$\mathbf{C} = \frac{1}{\det A} \begin{pmatrix} (\nabla g^\lambda)_\lambda \Lambda'_0 - (\nabla g^\Lambda)_\lambda \lambda'_0 \\ -(\nabla g^\lambda)_\Lambda \Lambda'_0 + (\nabla g^\Lambda)_\Lambda \lambda'_0 \end{pmatrix}.$$

(76) :

$$\left\{ \begin{array}{l} x'_1(\tau) = \left((\nabla g^x)_x + ((\nabla g^x)_\Lambda M_{11} + (\nabla g^x)_\lambda M_{21}) \right) x_1 + \\ \quad + \left((\nabla g^x)_y + ((\nabla g^x)_\Lambda M_{12} + (\nabla g^x)_\lambda M_{22}) \right) y_1 + \\ \quad + \left(\begin{pmatrix} (\nabla g^x)_\Lambda \\ (\nabla g^x)_\lambda \end{pmatrix}, \mathbf{C} \right), \\ y'_1(\tau) = \left((\nabla g^y)_x + ((\nabla g^y)_\Lambda M_{11} + (\nabla g^y)_\lambda M_{21}) \right) x_1 + \\ \quad + \left((\nabla g^y)_y + ((\nabla g^y)_\Lambda M_{12} + (\nabla g^y)_\lambda M_{22}) \right) y_1 + \\ \quad + \left(\begin{pmatrix} (\nabla g^y)_\Lambda \\ (\nabla g^y)_\lambda \end{pmatrix}, \mathbf{C} \right). \end{array} \right. \quad (77)$$

$$(\cdot, \cdot) = \mathbb{R}^n.$$

$$, \quad , \quad , \quad (38) \quad , \quad ,$$

$$\Lambda_0 = 0 \quad \Rightarrow \quad \mathbf{C} = \begin{pmatrix} \frac{\lambda'_0}{\alpha} \\ 0 \end{pmatrix}$$

$$, \quad g^x, g^y \quad \Lambda, \quad , \quad (77) :$$

$$\begin{pmatrix} (\nabla g^x)_\Lambda \\ (\nabla g^x)_\lambda \end{pmatrix} = \begin{pmatrix} 0 \\ * \end{pmatrix}, \quad \begin{pmatrix} (\nabla g^y)_\Lambda \\ (\nabla g^y)_\lambda \end{pmatrix} = \begin{pmatrix} 0 \\ * \end{pmatrix}.$$

(77) :

$$\left(\begin{pmatrix} (\nabla g^x)_\Lambda \\ (\nabla g^x)_\lambda \end{pmatrix}, \mathbf{C} \right) = \left(\begin{pmatrix} (\nabla g^y)_\Lambda \\ (\nabla g^y)_\lambda \end{pmatrix}, \mathbf{C} \right) = 0.$$

$$, \quad (38) \quad (x_1, y_1) \quad .$$

2) (3.3). x_0 .

x_0 , , Λ_0, λ_0 x_0, y_0 :

$$\begin{aligned} x_0'' &= (\nabla g^x)_x x_0' + (\nabla g^x)_y y_0' + (\nabla g^x)_\Lambda \Lambda_0' + (\nabla g^x)_\lambda \lambda_0' = \\ &= \left((\nabla g^x)_x + (\nabla g^x)_\Lambda \frac{\partial \Lambda_0}{\partial x} + (\nabla g^x)_\lambda \frac{\partial \lambda_0}{\partial x} \right) x_0' + \\ &\quad + \left((\nabla g^x)_y + (\nabla g^x)_\Lambda \frac{\partial \Lambda_0}{\partial y} + (\nabla g^x)_\lambda \frac{\partial \lambda_0}{\partial y} \right) y_0'. \end{aligned} \quad (78)$$

, $\Lambda_0(x_0, y_0), \lambda_0(x_0, y_0)$:

$$\begin{cases} 0 = g^\Lambda(\mathbf{X}_0), \\ 0 = g^\lambda(\mathbf{X}_0). \end{cases} \quad (79)$$

x :

$$\begin{cases} (\nabla g^\Lambda)_x + (\nabla g^\Lambda)_y \underbrace{y_x}_0 + (\nabla g^\Lambda)_\Lambda \cdot \Lambda_x + (\nabla g^\Lambda)_\lambda \cdot \lambda_x = 0, \\ (\nabla g^\lambda)_x + (\nabla g^\lambda)_y \underbrace{y_x}_0 + (\nabla g^\lambda)_\Lambda \cdot \Lambda_x + (\nabla g^\lambda)_\lambda \cdot \lambda_x = 0. \end{cases}$$

:

$$\begin{pmatrix} (\nabla g^\Lambda)_\Lambda & (\nabla g^\Lambda)_\lambda \\ (\nabla g^\lambda)_\Lambda & (\nabla g^\lambda)_\lambda \end{pmatrix} \begin{pmatrix} \Lambda_x \\ \lambda_x \end{pmatrix} = - \begin{pmatrix} (\nabla g^\Lambda)_x \\ (\nabla g^\lambda)_x \end{pmatrix}.$$

(Λ_x, λ_x) :

$$\begin{aligned} \Lambda_x &= \frac{(\nabla g^\Lambda)_\lambda (\nabla g^\lambda)_x - (\nabla g^\Lambda)_x (\nabla g^\lambda)_\lambda}{(\nabla g^\Lambda)_\Lambda (\nabla g^\lambda)_\lambda - (\nabla g^\Lambda)_\lambda (\nabla g^\lambda)_\Lambda}, \\ \lambda_x &= \frac{(\nabla g^\Lambda)_x (\nabla g^\lambda)_\Lambda - (\nabla g^\Lambda)_\Lambda (\nabla g^\lambda)_x}{(\nabla g^\Lambda)_\Lambda (\nabla g^\lambda)_\lambda - (\nabla g^\Lambda)_\lambda (\nabla g^\lambda)_\Lambda}. \end{aligned}$$

, M :

$$\frac{\partial \Lambda_0}{\partial x} = M_{11}, \quad \frac{\partial \lambda_0}{\partial x} = M_{21}, \quad \frac{\partial \Lambda_0}{\partial y} = M_{12}, \quad \frac{\partial \lambda_0}{\partial y} = M_{22}.$$

(78) :

$$\begin{aligned} (x_0')' &= (\nabla g^x)_x x_0' + (\nabla g^x)_y y_0' + (\nabla g^x)_\Lambda \Lambda_0' + (\nabla g^x)_\lambda \lambda_0' = \\ &= ((\nabla g^x)_x + (\nabla g^x)_\Lambda M_{11} + (\nabla g^x)_\lambda M_{21}) x_0' + \\ &\quad + ((\nabla g^x)_y + (\nabla g^x)_\Lambda M_{12} + (\nabla g^x)_\lambda M_{22}) y_0'. \end{aligned} \quad (80)$$

, (38) (77) , (80) (77) . ■

2. G_k^ρ :

$$G_{k+1}^\rho = \frac{1}{2} \sum_{m=1}^k \mathfrak{D}_m \mathfrak{D}_{k+1-m} g^\rho + \sum_{m=1}^k \mathfrak{D}_m \hat{G}_{k+1-m}^\rho,$$

$\hat{G}_{k+1}^\rho$ G_{k+1}^ρ :

$$G_k^\rho = \sum_{j=2}^k \frac{\mathcal{L}_k^{(j)} g^\rho}{j!}, \quad \hat{G}_k^\rho = \sum_{j=2}^k \frac{\mathcal{L}_k^{(j)} g^\rho}{(j+1)!}, \quad \rho \in \lambda, \Lambda, x, y.$$

:

$$\begin{aligned}
\mathcal{L}_{k+1}^{(j)} g^\rho &= \sum_{m=1}^{k+2-j} \mathfrak{D}_m \mathcal{L}_{k+1-m}^{(j-1)} g^\rho \\
G_{k+1}^\rho &= \sum_{j=2}^{k+1} \frac{\mathcal{L}_{k+1}^{(j)} g^\rho}{j!} = \sum_{j=2}^{k+1} \sum_{m=1}^{k+2-j} \frac{\mathcal{L}_{k+1-m}^{(j-1)} g^\rho}{j!} = \sum_{m=1}^k \mathfrak{D}_m \sum_{j=2}^{k+2-m} \frac{\mathcal{L}_{k+1-m}^{(j-1)} g^\rho}{j!} = \\
&= \sum_{m=1}^k \mathfrak{D}_m \sum_{l=1}^{k+1-m} \frac{\mathcal{L}_{k+1-m}^{(l)} g^\rho}{(l+1)!} = \sum_{m=1}^k \mathfrak{D}_m \left(\sum_{l=2}^{k+1-m} \frac{\mathcal{L}_{k+1-m}^{(l)} g^\rho}{(l+1)!} + \frac{1}{2} \mathcal{L}_{k+1-m}^{(1)} g^\rho \right) = \\
&= \sum_{m=1}^k \mathfrak{D}_m \left(\hat{G}_{k+1-m}^\rho + \frac{1}{2} \mathfrak{D}_{k+1-m} g^\rho \right) \blacksquare
\end{aligned}$$

3. 1) $\lambda_j, \Lambda_j, x_j, y_j, \quad j \leq k-1 \quad \tau \quad ,$
 $G_k^x(\tau), \tilde{G}_k^x(\tau), G_k^\Lambda(\tau), G_k^y(\tau) \quad . \quad G_k^x(\tau), \tilde{G}_k^x(\tau), G_k^\Lambda(\tau) \quad , \quad G_k^y(\tau), \tilde{G}_k^y(\tau) \quad G_k^x(\tau) \quad .$

2) $G_2^x(\tau), \tilde{G}_2^x(\tau), G_2^\Lambda \quad , \quad G_2^y(\tau), \tilde{G}_2^y(\tau) \quad .$

:

1) $\dots x_0(\tau) \quad , \quad y_0(\tau), \lambda_0(\tau) \quad - \quad :$

$$g^\Lambda|_{\lambda_0(\tau), \Lambda_0(\tau), x_0(\tau), y_0(\tau)} = U(x_0, y_0) \sin \lambda_0 - V(x_0, y_0) \cos \lambda_0 \quad - \quad ,$$

$$g^x|_{\lambda_0(\tau), \Lambda_0(\tau), x_0(\tau), y_0(\tau)} = -\varepsilon \left(2F y_0 - \frac{\partial U}{\partial y} \cos \lambda_0 - \frac{\partial V}{\partial y} \sin \lambda_0 \right) \quad - \quad ,$$

$$g^y|_{\lambda_0(\tau), \Lambda_0(\tau), x_0(\tau), y_0(\tau)} = \varepsilon \left(2F(x_0 + \hat{x}_0) + G - \frac{\partial U}{\partial x} \cos \lambda_0 - \frac{\partial V}{\partial x} \sin \lambda_0 \right) \quad - \quad .$$

:

- $(x_0) \quad .$
- $(y_0 \quad \lambda_0) \quad .$

, $+1 \quad -1 \quad - \quad , \quad :$

$$\begin{aligned}
\frac{\partial^{p+q+m} g^\Lambda}{\partial x^p \partial y^q \partial \lambda^m} \Big|_{\lambda_0(\tau), \Lambda_0(\tau), x_0(\tau), y_0(\tau)} &\Leftrightarrow \quad -(-1)^{q+m}, \\
\frac{\partial^{p+q+m} g^x}{\partial x^p \partial y^q \partial \lambda^m} \Big|_{\lambda_0(\tau), \Lambda_0(\tau), x_0(\tau), y_0(\tau)} &\Leftrightarrow \quad -(-1)^{q+m}, \\
\frac{\partial^{p+q+m} g^y}{\partial x^p \partial y^q \partial \lambda^m} \Big|_{\lambda_0(\tau), \Lambda_0(\tau), x_0(\tau), y_0(\tau)} &\Leftrightarrow \quad (-1)^{q+m}.
\end{aligned} \tag{81}$$

, $G_k^\Lambda(\tau), G_k^x(\tau), G_k^y(\tau) \quad :$

$$G_k^\Lambda(\tau) = \sum_{p_1, q_1, m_1, \dots} \frac{\partial^{p+q+m} g^\Lambda}{\partial x^p \partial y^q \partial \lambda^m} \Big|_{\lambda_0(\tau), \Lambda_0(\tau), x_0(\tau), y_0(\tau)} (x_{p_1} y_{q_1} \lambda_{m_1} \dots),$$

$$G_k^x(\tau) = \sum_{p_1, q_1, m_1, \dots} \frac{\partial^{p+q+m} g^x}{\partial x^p \partial y^q \partial \lambda^m} \Big|_{\lambda_0(\tau), \Lambda_0(\tau), x_0(\tau), y_0(\tau)} (x_{p_1} y_{q_1} \lambda_{m_1} \dots),$$

$$G_k^\Lambda(\tau), G_k^x(\tau), G_k^y(\tau), \quad g^\Lambda, g^x, g^y, \quad \tilde{G}_k^x, \tilde{G}_k^y, \tilde{G}_k^\Lambda(\tau), \tilde{G}_k^x(\tau), \tilde{G}_k^y(\tau).$$

$$G_2^\rho(\tau) = \sum_{\substack{p+q+m=2, \\ p,q,m \geq 0, \\ p,q,m \neq 2}} \frac{\partial^2 g^\rho}{\partial x^p \partial y^q \partial \lambda^m} \Big|_{\lambda_0(\tau), \Lambda_0(\tau), x_0(\tau), y_0(\tau)} x_1^p y_1^q \lambda_1^m \quad (82)$$

$$, x_1^p y_1^q \lambda_1^m \quad (-1)^p \cdot, \quad (81), \quad (82) \quad -(-1)^{p+q+m} \quad \rho \in \{\Lambda, x\} \quad (-1)^{p+q+m} \quad \rho = y. \quad p+q+m=2, :$$